PORTFOLIO MANAGEMENT

4

DIVERSIFY TO REDUCE RISK

A unit trust represents the pooling of investors' funds into a collective investment. Investment managers such as Allan Gray, Old Mutual, Sanlam, and Coronation invest these funds in shares, property, bonds and money market instruments. Unit trusts enable small investors to invest a small amount of money and yet achieve effective diversification. According to Allan Gray, the benefit of diversification is that risk is spread among investments that are uncorrelated.

Vuyo Nogantshi at **Allan Gray** *wrote that "the best way to reduce your concentration risk, without losing out on the potential to earn good returns, is to make sure that you are invested in a combination of assets that have little correlation to one another – essentially, having a diversified portfolio where you generate returns from a wider spread of assets, industries and markets with an acceptable level of risk" (GrayIssue, 30 May 2019). Offshore equity investing will improve diversification and enable South Africans to invest in companies such as Amazon, Apple and Microsoft. South Africans will also be able to invest in sectors that are poorly represented on the JSE such as information technology, healthcare, pharmaceuticals and biotechnology. It is also important to realise just how concentrated the JSE is, as ten companies make up about 60% of the JSE ALSI. Avoiding concentration and achieving a greater degree of diversification is a good reason to invest offshore. Diversification also involves investing across asset classes such as shares, bonds and other assets. Yet one could argue that in relation to the MSCI World Index with 1512 companies, just seven companies (Apple, Microsoft, Amazon, Nvidia, Alphabet, Tesla, and Meta) make up close to 20% of the market capitalisation of the MSCI World Index on 31 July 2023. Therefore, concentration risk is a factor to consider even when investing in the MSCI World Index. The* **Allan Gray Balanced Fund** *states in its July 2023 fact sheet that "the Fund invests in a mix of shares, bonds, property, commodities and cash. The Fund can invest a maximum of 45% offshore." In this chapter we will explore the concept of diversification and portfolio theory.*

LEARNING OBJECTIVES

After working through this chapter, you should be able to:

- Explain the impact of diversification on the expected return and risk of a portfolio of shares.
- Calculate the following indicators of risk and expected return on a portfolio of shares:
  - expected return on a two-asset portfolio;
  - risk of a two-asset portfolio using covariance;
  - expected return on a multi-share portfolio;
  - beta of a portfolio;
  - expected return on a leveraged portfolio;
  - standard deviation of a leveraged portfolio.
- Understand the concept of an efficient frontier.
- Apply the concept of the security market line.
- Evaluate the concept of market efficiency.
- Understand how Exchange Traded Funds (ETFs) work.
- Explain how behavioural finance may influence portfolio decisions.
- Use Excel to determine betas.
- Understand the use of betas in practice.

Page 4-2 INTRODUCTION

The old adage that you shouldn't place all your eggs in one basket reveals a general approach to risk. Because virtually all future outcomes are uncertain, it would not be wise for an investor to place all available funds in one investment. At the time of investment, it is not known with certainty which investments will succeed and which will fail. It is therefore sensible to diversify into a number of investments in the expectation that those which are profitable will at least compensate for the loss which

may be sustained from those that are not. This is the fundamental principle for adding a risk premium to the risk-free rate, for an individual investment opportunity which is not risk-free, in order to arrive at the required return for that investment. This principle was fully outlined in Chapter 3.

In this chapter we describe the effect on expected return and estimated risk of diversifying – of investing in more than a single investment. We confine the discussion to investments on securities markets, because the share prices are known at the time of purchase and there is a past record of returns available which is useful for forecasting future returns. The principles of portfolio management may, however, be generalised to encompass all assets in which an investor may invest.

Using modern portfolio theory and the capital asset pricing model, it will be demonstrated that holding a portfolio of shares will always be a superior investment strategy to selecting an individual share. Portfolio management is based on the work of many researchers but pioneered by Markowitz (1959) and Sharpe (1964). Portfolio theory is based on a number of assumptions regarding the way in which investors and stock markets function. While these assumptions may seem restrictive, the principles are fundamental to an understanding of investment strategies and are essential to an understanding of financial principles.

It also implies that if investors can effectively diversify risk by investing in a portfolio of assets such as shares, then a firm does not need to diversify its operating assets in order to reduce the investment risk to its shareholders.

1 TWO-ASSET PORTFOLIO RISK AND RETURN

Holding more than one investment in financial assets is generally referred to as holding a *portfolio of investments* (although there may be a one-asset portfolio). The principles used when establishing the expected return of a portfolio will be discussed using a two-asset portfolio and then by generalising the principles to a portfolio with many financial assets.

Example 4.1: Investing in a two-asset portfolio

An investor decides to invest R10 000 in a portfolio which may consist of any combination of shares in Plasco Ltd and Quinco Ltd. The following summary expected return (R_e) and standard deviation (σ) statistics are available for the two companies:

Plasco $R_e = 30\%$; $\sigma = 12.6\%$ and Quinco $R_e = 20\%$; $\sigma = 10.3\%$

If, for example, R5 000 is invested in Plasco and R5 000 in Quinco, it would be expected that the return on the portfolio would be a weighted average of the proportion invested in each. This example will be used as a basis for establishing the measure of return and the measure of risk for possible portfolio combinations.

Measuring two-asset portfolio returns

An investor who decides to diversify by buying shares in two companies will expect a return dependent on the proportion of funds invested in each share. The expected return of the portfolio will quite simply be the weighted average of the expected returns of the individual shares as depicted in Table 4.1. Page 4-3

Table 4.1 Expected returns for various portfolios of *P* and *Q*

	A	B	C
1	% Investment in P Ltd Return = 30%	% Investment in Q Ltd Return = 20%	Expected return on the portfolio
2	0%	100%	20.0%
3	25%	75%	22.5%
4	50%	50%	25.0%
5	75%	25%	27.5%
6	100%	0%	30.0%
7	Example of Workings: (25% x 30%) + (75% x 20%) = 22.5%		

It is clear from Table 4.1 that the greater the proportion of funds invested in P Ltd, the higher the expected return becomes. The general formula for calculating the expected return of a portfolio is:

$$R_p = \sum_{i=1}^{n} W_i \times R_i \qquad \textit{(Formula 4.1)}$$

where: R_p = the expected return on the portfolio

W_i = the proportion of funds invested in share *i*

R_i = the expected return on share *i*

Expected return is not the only parameter on which to base the decision. A measure of risk for the various portfolio combinations is required.

The principles of portfolio risk

The risk of a portfolio depends not only on the riskiness of the individual shares which compose the portfolio, but also on the relationship between their returns. The tendency may be to expect that portfolio risk is calculated in a similar fashion to portfolio return – i.e. using a weighted average. However, this is not the correct procedure.

Assume we are able to collect the historical information on the share price performance of the two shares P and Q, and they are as listed in Table 4.2.

Table 4.2 Past performance of shares P and Q, and an equally weighted portfolio PQ

	A	B	C	D	E	F
1				Percentage returns		
2	Year		Note	P	Q	PQ
3	1			40%	32%	36%
4	2			16%	8%	12%
5	3			44%	30%	37%
6	4			20%	14%	17%
7	5			44%	30%	37%
8	6			16%	6%	11%
9	7			42%	28%	35%
10	8			18%	12%	15%
11	*Average return*		1	**30%**	**20%**	**25%**
12	Std. Dev.		2	**12.6%**	**10.3%**	**11.4%**
13	Covariance of P & Q		3			**0.0127**
14	Correlation Coefficient		4			**0.9783**
15	**Note: Excel Functions**					
16	1	=AVERAGE(D3:D10)				
17	2	=STDEVP(D3:D10)				
18	3	=COVAR(D3:D10,E3:E10)				
19	4	=CORREL(D3:D10,E3:E10)				

Page 4-4 The following is apparent from the data collected and processed:

- Share P has an average return of 30% over the period, and share Q has an average return of 20%.
- The standard deviation of returns has been calculated using Excel, and reflects the greater variability of returns for share P, with a standard deviation of 12.6% compared to 10.3% for share Q.
- Of particular relevance is the fact that the two shares tend to move in similar directions in each year. In year 1, for example, both shares achieved well above the mean return, whereas in year 2 both shares achieved well below their mean. This pattern is evident throughout the 8-year period.

- The standard deviation of an equally weighted portfolio lies between those of each individual share. Because the shares move in near-perfect unison, the risk of the portfolio reflected in the standard deviation of 11.4% is very close to the weighted average of the two standard deviations.

The results are reflected in Figure 4.1.

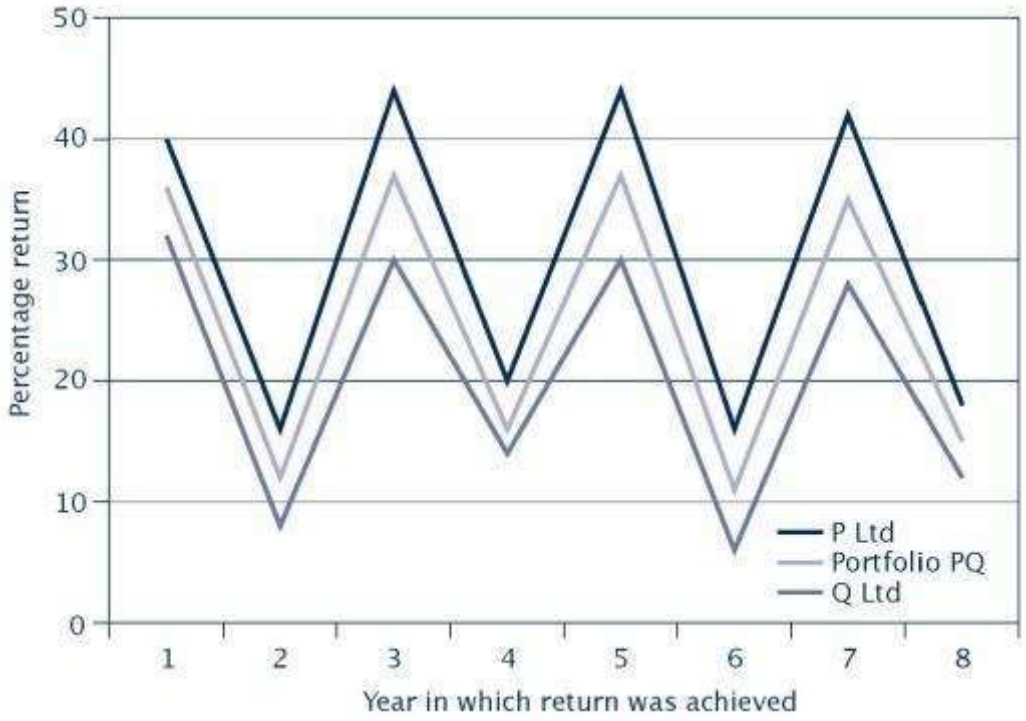


Figure 4.1 Returns on P and Q: near-perfect unison: 50/50 portfolio

The measure of harmony in movement between the two shares can be calculated in further summary statistics. When only one share was under consideration we used the square root of its variance, the *standard deviation*. For two shares we use the *covariance*, often expressed in a standardised form as the *correlation co-efficient*. This statistic reflects the degree to which two variables move together. The values of correlation lie between +1, which is perfect positive correlation, and –1, which is perfect negative correlation. Note the correlation coefficient is a high positive figure of 0.978.

A correlation co-efficient of 0 indicates that there is no correlation between the two variables. In this hypothetical example, where P and Q move in near-perfect unison, a state of near-perfect positive correlation exists. This situation would seldom exist in practice.

Covariance and correlation are explained in greater detail in Chapter 3 and the data set out in Table 4.1 for shares P and Q is used in Chapter 3 to define and determine the covariance and correlation co-efficient. We apply the appropriate calculations, as well as use Excel to determine covariance and correlation. Please refer to Chapter 3. Page 4-5

Assume now that all the rates of return are identical to those in Table 4.2, but that the timing is different. Notice from Table 4.3 that the average return and standard deviation for each share remains identical, but the co-movement, reflected in the covariance statistic and the correlation co-efficient, is significantly different.

Table 4.3 Past performance of shares P and Q, and an equally weighted portfolio PQ

	A	B	C	D	E	F
21				Percentage returns		
22	Year			P	Q	PQ
23	*1*			40%	12%	26%
24	*2*			16%	32%	24%
25	*3*			44%	8%	26%
26	*4*			20%	30%	25%
27	*5*			44%	14%	29%
28	*6*			16%	30%	23%
29	*7*			42%	6%	24%
30	*8*			18%	28%	23%
31	*Average return*			**30%**	**20%**	**25%**
32	Std. Dev.			**12.6%**	**10.3%**	**1.9%**
33	Covariance of P & Q					**-0.0126**
34	Correlation Coefficient					**-0.9667**

If the data were as presented in Table 4.3:

- Share P has an average return of 30% over the period, and share Q has an average return of 20%. This is identical to the previous case.
- The standard deviation of returns has been calculated using Excel (= STDEVP), and reflects the greater variability of returns for share P, with a standard deviation of 12.6% compared to 10.3% for share Q. This too is identical to the first case. Note that we have determined the population standard deviation. We did not use the STDEV function, which calculates the standard deviation for a sample[1].
- The difference, however, is that the two shares move in opposite directions in each year. In Year 1, for example, when P performs well above its mean return, Q

performs well below its mean return. This pattern is evident throughout the 8-year period.

- The result of this pattern of returns is that an equally weighted portfolio tends to have very similar returns each year, thus there is much greater consistency, and less variability, reflected in the very small standard deviation of the portfolio of only 1.9%.
- Of equal interest is the correlation co-efficient, which is very highly negative at minus 0.9667.

Later in the chapter, the manual calculation of these important summary statistics for a two- asset portfolio will be demonstrated. Figure 4.2 illustrates the movement of each share over the 8-year period, and that of an equally weighted portfolio, based on the historical returns in Table 4.3. Page 4-6

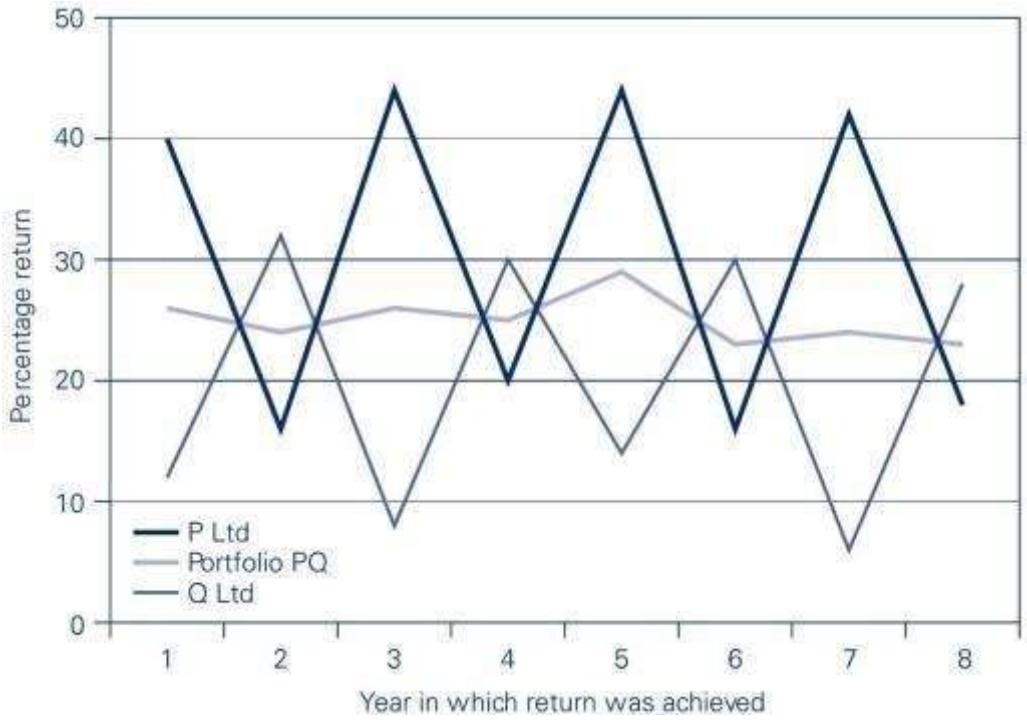


Figure 4.2 Returns on P and Q: near-perfect negative correlation: 50/50 portfolio

The frequency distributions of returns on share P, share Q, and the portfolio of shares P and Q respectively, are roughly graphed in Figure 4.3.

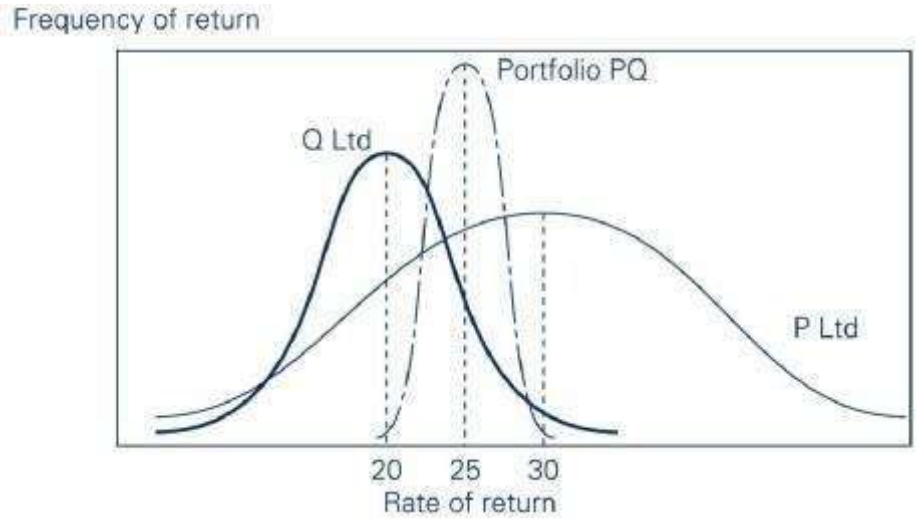


Figure 4.3 Frequency distribution of two individual shares and the portfolio of two shares

The following can be noted:

- The dispersion around the mean of Q Ltd is considerably tighter than the dispersion around the mean of P Ltd. This is in accordance with the expectation that P Ltd is more risky than Q Ltd, but offers a higher expected return.
- The dispersion around the mean of the portfolio of 50% P Ltd and 50% Q Ltd is tighter than that of either P Ltd or Q Ltd. This means that the risk of the portfolio, in this case, is lower than the risk of holding either P Ltd or Q Ltd individually.

Just as it is unlikely that two shares will be perfectly positively correlated, it is also unlikely that two shares will be perfectly negatively correlated. Because of the many general factors that are likely to influence all shares in a fairly similar manner at any given time, shares will tend to be positively correlated rather than negatively correlated. In general, however, it can be accepted that anything less than perfect positive correlation will cause the portfolio risk to be lower than the weighted average of the standard deviations of the two shares held in the portfolio.

Measuring two-asset portfolio risk

The risk of a two-asset portfolio has been demonstrated in principle to depend upon the extent to which returns move in harmony with each other. The extent of this co-movement may be measured by calculating the covariance between two shares, denoted by $cov_{A,B}$, or its normalised statistic, the *correlation co-efficient*, denoted as ρAB. Once these indicators have been calculated, it is possible to derive the variance

and standard deviation of the two-asset portfolio. These procedures are best illustrated using an example.

Example 4.2: Statistical calculations for two-asset portfolio

Information is available for two shares – Benix Ltd, listed under the general retail sector, and Genhold Ltd, listed under the media and entertainment sector. The returns for shareholders have been calculated for the last five years.

Year	Benix	Genhold
1	26%	24%
2	20%	35%
3	22%	22%
4	23%	37%
5	29%	32%

Before proceeding with the statistical procedures illustrated in Table 4.4, the following must be noted:

- The number of data points used is for illustrative purposes only. Five points are inadequate for any meaningful statistical interpretation in practice.
- It is assumed that the distribution has the characteristics of a standard normal distribution.
- Past performance is considered to be valid for the purpose of predicting future expectations.

Table 4.4 provides the basis for calculating the mean and standard deviation for each share, as well as the covariance and correlation co-efficient of the two shares.

Table 4.4 Calculation of the mean, standard deviation and covariance

	1	2	3	4	5	6	7
Year	**R_b**	**R_g**	**d_b**	**^{d}g**	**d_b^2**	**d_a^2**	**$d_b x d_g$**
1	26.00%	24.00%	2.00%	-6.00%	0.04%	0.36%	-0.12%
2	20.00%	35.00%	-4.00%	5.00%	0.16%	0.25%	-0.20%
3	22.00%	22.00%	-2.00%	-8.00%	0.04%	0.64%	0.16%
4	23.00%	37.00%	-1.00%	7.00%	0.01%	0.49%	-0.07%
5	29.00%	32.00%	5.00%	2.00%	0.25%	0.04%	0.10%
Sum					0.50%	1.78%	-0.13%
Avg return	**24.00%**	**30.00%**					
Variance				**= $d^2/5$**	**0.10%**	**0.356%**	
Std Dev.				**= $Var^{0.5}$**	**3.16%**	**5.97%**	
Cov_{bg}				**$[d_b x d_g]/5$**			**-0.026%**

From Table 4.4 the following should be noted:

- Columns one and two list the past returns to shareholders over the last five years. Each column is totalled and divided by the numbers of readings (five) in order to obtain the mean (x), which then serves as the expected return.
- Columns three and four reflect the deviations from the respective means. These are squared in columns five and six. The columns are totalled and divided by the number of readings (five) in order to obtain the variances (σ^2) of 0.10% and 0.356%. The variances may be converted to standard deviations (σ) by finding their square roots. Page 4-8
- Each deviation of Benix (d_b) in column three is multiplied by the deviations of Genhold (d_g) in column four, in order to obtain (d_b)(d_g) in column seven. These are totalled and divided by the number of readings (five) in order to obtain the covariance cov_{bg} of – 0.026%.
- Note that these two shares did not always move in harmony. For example, in year one, when Benix achieved 2% above its mean performance of 24%, Genhold achieved 6% below its mean performance of 30%. These contrary performances have resulted in a negative covariance of –0.026%.
- The covariance term may be normalised – i.e. expressed in a manner such that all readings will fall between –1 and +1. Such normalised covariance is called the correlation co-efficient and is calculated as follows:

$$\rho_{bg} = \frac{cov_{bg}}{\sigma_b \times \sigma_g} \qquad \textit{(Formula 4.2)}$$

$$= \frac{-0.026\%}{3.16\% \times 5.97\%}$$

$$= -0.14$$

where: ρ_{bg} = correlation co-efficient of B and G

σ_b = standard deviation of B

σ_g = standard deviation of G

cov_{bg} = covariance of B and G

This summary statistic indicates that there is a slight negative correlation between Benix and Genhold.

It also follows from Formula 4.2 that if both standard deviations are known, as well as the correlation co-efficient, then the covariance may be calculated as follows, all notation being identical to Formula 4.2

$$\text{cov}_{bg} = \rho_{bg} \times \sigma_b \times \sigma_g \qquad \textit{(Formula 4.3)}$$

We now wish to measure the variance of a portfolio of shares comprising a proportion of both Benix and Genhold shares. We will thus develop a variance matrix from which a formula for calculating the variance and standard deviation of a portfolio may be derived.

Assume we wish to construct a portfolio of 25% Benix and 75% Genhold. The first step in constructing a variance matrix requires that the relative weighting for each block in the matrix be determined. This is illustrated in Table 4.5.

Table 4.5 Weights in a variance matrix. This requires that the relative weighting for each block in the matrix be determined.

	Benix (25%)	Genhold (75%)
Benix (25%)	0.0625	0.1875
Genhold (75%)	0.1875	0.5625

From Table 4.5, the following must be noted:

- The figures obtained in each block represent the weighting which each will have in the total variance of the portfolio.
- The sum of the weightings is equal to one.

The relative variance and covariance figures may now be inserted. In Table 4.6, it is apparent, for example, that the block matching Benix with Benix uses the variance of Benix, while the block matching Benix with Genhold uses the covariance between Benix and Genhold.

Table 4.6 Variance matrix for Benix and Genhold

	Benix (25%)	Genhold (75%)
Benix (25%)	0.0625 × 0.10%	0.1875 × –0.026%
Genhold (75%)	0.1875 × –0.026%	0.5625 × 0.356%

The sum of all the variances and covariances multiplied by their weightings in the matrix equals 0.19675% (0.00625% + 0.20025% – 0.004875% – 0.004875%). This is the variance of the portfolio comprising 25% Benix and 75% Genhold.

This method of arriving at the variances is cumbersome and may be expressed and calculated by a formula which derives from the matrix as follows:

$$\sigma_p^2 = W_b^2 \sigma_b^2 + W_g^2 \sigma_g^2 + 2W_b W_g \times cov_{bg} \qquad \textit{(Formula 4.4)}$$

where: σ_p^2 = the variance of the portfolio

W_b^2 and W_g^2 = the weight (proportion) invested in B and G respectively (note: $W_b + W_g = 1$)

σ_b^2 and σ_g^2 = the variance of returns on shares B and G respectively

cov_{bg} = the covariance of the returns on the two shares

Applying Formula 4.4 to calculate the risk of a portfolio comprising 25% in Benix and 75% in Genhold results in the following:

$$\sigma_p^2 = (0.25)(0.25)(0.10\%) + (0.75)(0.75)(0.356\%) + (2)(0.25)(0.75)(-0.026\%)$$
$$= 0.19675\%$$

$$\text{So } \sigma_p = \sqrt{0.19675\%}$$
$$= 4.44\%$$

Note that Formula 4.4 can also be expressed using the correlation co-efficient, if the covariance is not known, as follows (with notation identical to Formula 4.4). Refer back to Formula 4.3 for a breakdown.

$$\sigma_p^2 = W_b^2 \sigma_b^2 + W_g^2 \sigma_g^2 + 2W_b W_g \rho_{bg} \sigma_b \sigma_g \qquad \textit{(Formula 4.5)}$$

Of course, you may not like Formulae 4.4 and 4.5. Well, in this example we can determine the standard deviation of the portfolio in another way. Firstly, we will use Excel to determine the weighted average return for each year by multiplying the return for each security by its weighting in the portfolio to determine a weighted-average return for each year. For example, we would determine the weighted-average return for Year 2 to be 31.3% [(0.25 × 20%) + (0.75 × 35%)]. We then determine the average return and standard deviation of the portfolio returns. In this case, we have used the Excel function [= STDEVP(B16..F16)] to determine the standard deviation of 4.44%. This agrees with the result that we obtained by using Formula 4.4 and is presented in Table 4.7.

Table 4.7 Portfolio return and portfolio standard deviation

Table 4.7 Portfolio return and portfolio standard deviation

	A	B	C	D	E	F	G	H
15	Portfolio	Yr 1	Yr 2	Yr 3	Yr 4	Yr 5	Avg Return	Stdev (σ)
16	BG	24.5%	31.3%	22.0%	33.5%	31.3%	29%	4.44%
17	Portfolio = 25% of B and 75% of G							

Page 4-10 **Example 4.3: Risk of two-asset portfolios**

Share P has R_e = 30%; σ = 15% and share Q has R_e = 20% and σ = 10%. The correlation co-efficient for the returns of the two companies is known to be +0.4. Determine the risk of the portfolio for weights of p = 100%, 75%, 50%, 25% and 0%.

Only one calculation is shown as an example, where W_p = 25% (by default W_q must be 75%), the others being tabulated in Table 4.8. Use Formula 4.5 to check the standard deviations of the portfolio combinations (σ_p) tabulated in Table 4.8. To solve for R_p refer back to Formula 4.1:

$$\begin{aligned}\sigma_p^2 &= (0.0625)(0.0225) + (0.5625)(0.01) + 2(0.75)(0.25)(0.4)(0.15)(0.10)\\ &= (0.001406) + (0.005625) + (0.002250)\\ &= 0.009281\end{aligned}$$

$$\begin{aligned}\text{So } \sigma_p &= \sqrt{0.009281}\\ &= 0.096338 = 9.6\%\end{aligned}$$

Note that in this case the risk of the portfolio as measured by the standard deviation is lower than the risk of Q Ltd. In other words, holding the portfolio is less risky than the lowest-risk individual share. This is a special case and cannot be generalised. In all cases, however, the portfolio risk measured by the portfolio standard deviation will be less than the weighted- average standard deviations of the two shares. The full range of risk for holding different proportions of the shares is as follows.

Table 4.8 Portfolio risk and return

% Invested in P Ltd	% Invested in Q Ltd	σ p	R p
100%	0%	15.0%	30.0%
75%	25%	12.5%	27.5%
50%	50%	10.5%	25.0%
25%	75%	9.6%	22.5%
0%	100%	10.0%	20.0%

We are now in a position to graph the expected return and risk of portfolios comprising different proportions of shares in P Ltd and Q Ltd. This is displayed in Figure 4.4.

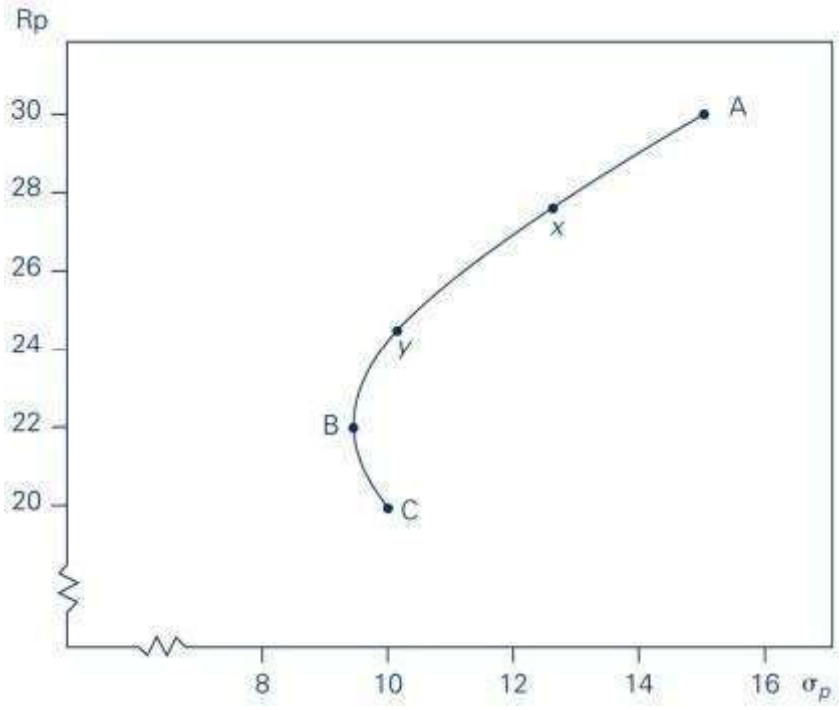


Figure 4.4 Feasible set of portfolio of two assets

Page 4-11 The line ABC represents a feasible set of portfolios, that is, all the possible combinations of P and Q are represented. It makes sense to hold a combination of P and Q which lies between points A and B, because any portfolio between B and C would be inferior to a portfolio between A and B. Point *y*, for example, is superior to point C because it results in a higher return for the same degree of risk. These relationships were originally identified by Markowitz, who gave the name *efficient frontier* to the line AB.

In Table 4.9, we have expanded Table 4.8 to include more portfolio weightings. We then apply Formula 4.5, assuming a correlation *(p*) of 0.4 and the same returns and standard deviations.

	P	Q
Return	30%	20%
Stdev	15%	10%

The resulting standard deviation and expected return is presented in Table 4.9. If the weighting of P is 25%, we will note that the portfolio standard deviation is 9.6%, which is what we calculated using Formula 4.5.

Table 4.9 Portfolio risk and return – expanded

Correlation =	0.4																
Weight P	0.00%	6.25%	12.50%	18.75%	25.00%	31.25%	37.50%	43.75%	50.00%	56.25%	62.50%	68.75%	75.00%	81.25%	87.50%	93.75%	100.00%
Weight Q	100.00%	93.75%	87.50%	81.25%	75.00%	68.75%	62.50%	56.25%	50.00%	43.75%	37.50%	31.25%	25.00%	18.75%	12.50%	6.25%	0.00%
Formula 4.5																	
$(\text{Stdev PQ})^2$ =	0.010	0.010	0.009	0.009	0.009	0.010	0.010	0.010	0.011	0.012	0.013	0.014	0.016	0.017	0.019	0.021	0.023
Stdev PQ	0.100	0.098	0.097	0.096	**0.096**	0.097	0.099	0.102	0.105	0.109	0.114	0.119	0.125	0.131	0.137	0.143	0.150
Return PQ	0.200	0.206	0.213	0.219	0.225	0.231	0.238	0.244	0.250	0.256	0.263	0.269	0.275	0.281	0.288	0.294	0.300

We have then calculated the same tables assuming a correlation (p) of 1, –1 and 0. We can plot this to see how correlation will affect portfolio risk as measured by its standard deviation. The relationship between the portfolio return and risk for each level of correlation (p) is depicted in Figure 4.5.

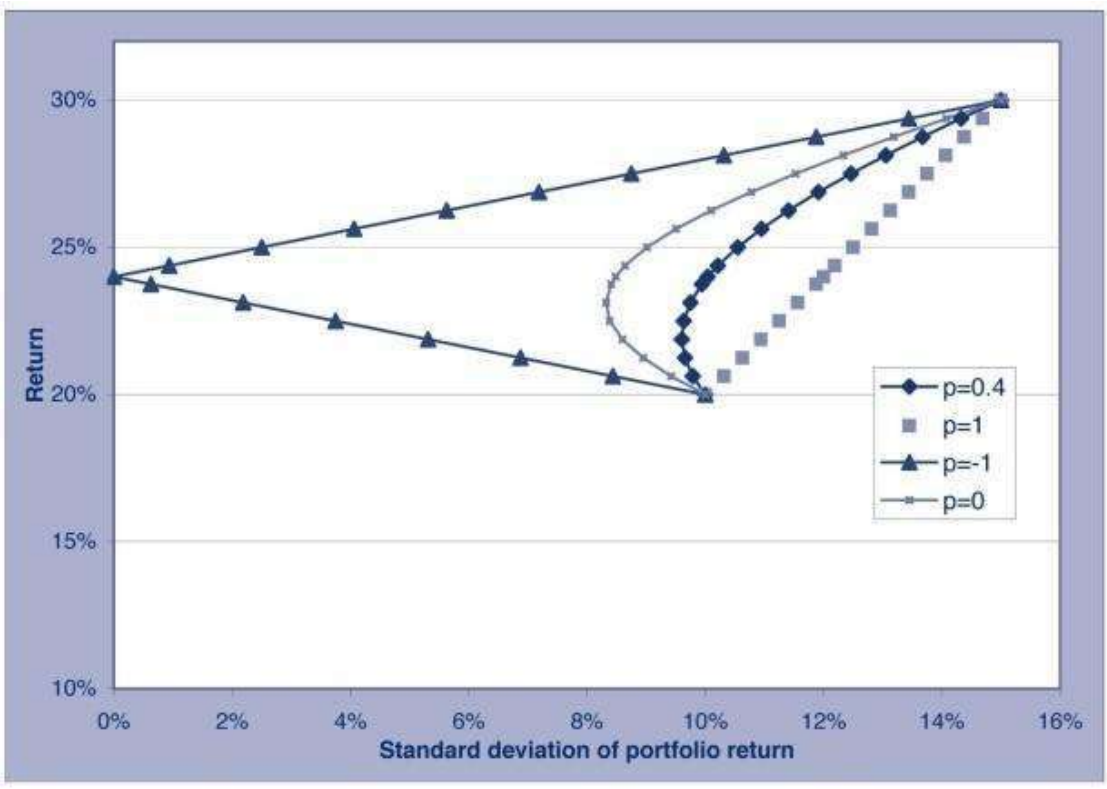


Figure 4.5 Portfolio risk and return for P & Q

Page 4-12 Perfect correlation (p = 1) results in a straight line from P to Q and so diversification does not result in lower risk relative to return as the standard deviation is simply a weighted average of the standard deviations of P and Q. If the correlation between the share returns of P and Q are less than 1, then there will be some benefit from diversification. If P and Q are perfectly negatively correlated (p = –1), then this will result in a significant reduction in risk depending on the relative weightings of P and Q in the portfolio. A portfolio made up of 40% of P and 60% of Q will result in a portfolio with zero risk, if the share returns are perfectly negatively correlated (p = –1).

It is important to note that with diversification and less than perfect correlation, we get a weighted-average rate of return but a less than weighted-average level of risk.

We get something for nothing.

Positioning an investor on the efficient frontier

Which combination of shares should an investor select? Unfortunately, there is no way of calculating this as it is dependent on the utility function of the investor. The term *utility* is derived from economics and refers to the subjective satisfaction obtained by an individual from taking certain actions or being subjected to certain circumstances. The well-known concept of diminishing marginal utility implies that, as expected investment return increases, the additional subjective satisfaction of an investor declines at an increasing rate. The rate of decline is dependent upon the attitude toward risk of the individual investor. The more risk-averse an individual is, the steeper the decline in marginal utility will be. This is illustrated in Figure 4.6.

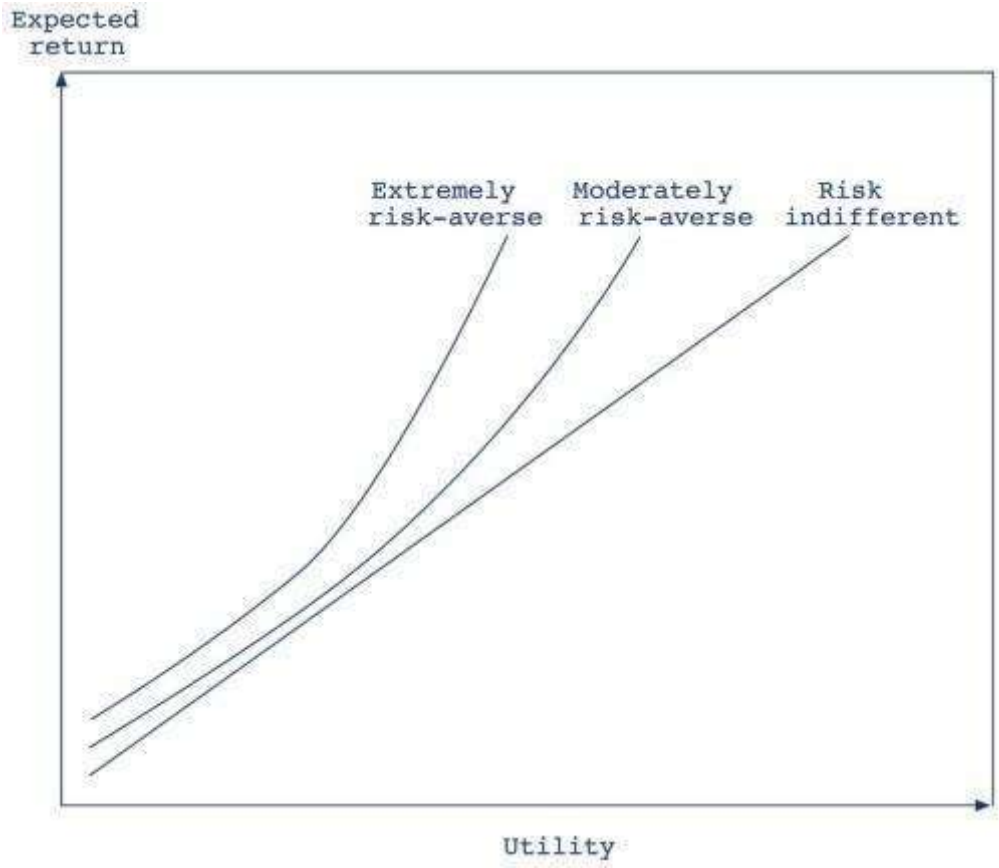


Figure 4.6 Utility curves

From Figure 4.6 it is clear that the risk-indifferent individual displays no diminishing marginal utility, i.e. he or she continues to derive utility proportional to the increase in expected investment returns. The utility function is, therefore, linear. The moderately risk- averse individual requires increasingly higher returns in order to enjoy a given level of utility – the curve therefore adopts a convex shape. The extremely risk-averse

investor's curve Page 4-13 slopes even more steeply upward, indicating the relatively greater increase in expected return required for a given quantum of utility.

The straight line and curves in Figure 4.6 are also known as *indifference* curves. This means that individuals represented by those curves are indifferent to investment opportunities offering the combination of return and utility lying along their particular slope. It is important to note that it is the slope, or steepness, of the line that distinguishes the risk characteristics of individual investors.

The utility functions can be superimposed onto the risk-return space illustrated in Figure 4.7. Figure 4.7 indicates the indifference curves for two investors X and Y.

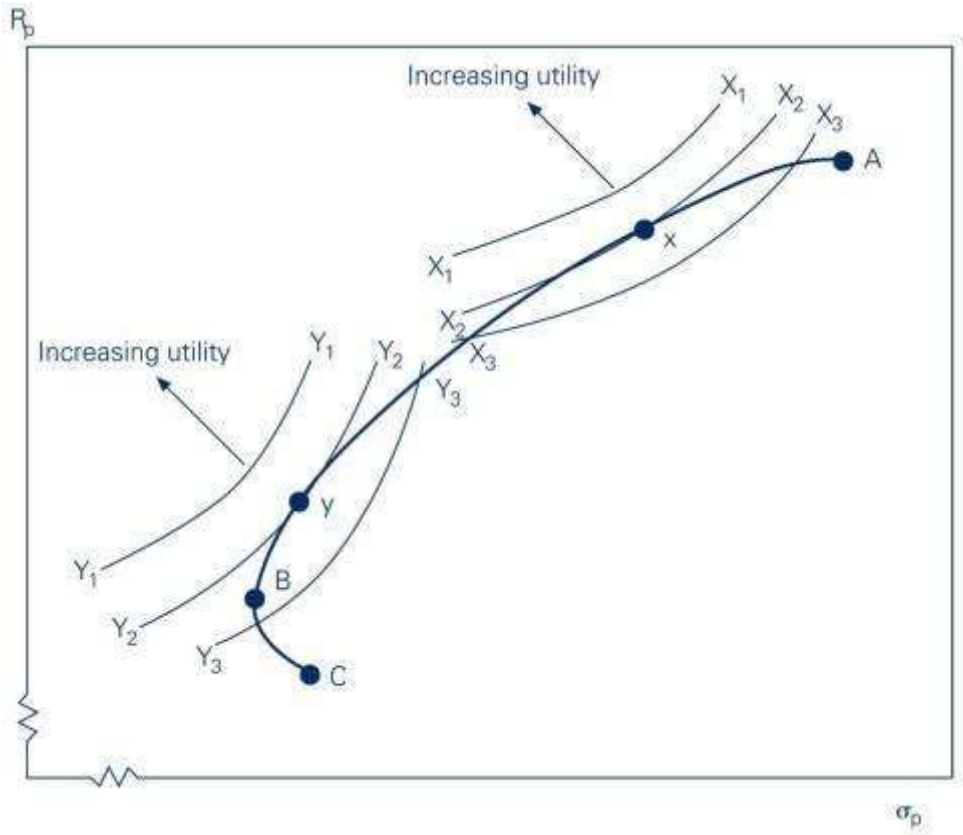


Figure 4.7 Indifference functions

The following can be seen in Figure 4.7:

- Three discrete indifference curves of investor X are X_1, X_2, and X_3. The curves proceed in an upward and leftward direction, depicting the increasing return being sought for increased risk. The slope of the curve reflects the risk preference of the investor. Curve X_1does not encounter any investment opportunity. Curve X_2, although offering less utility, is the first to contact the opportunity set of feasible portfolios on the efficient frontier. Curve X_3 meets the efficient frontier at two

points, although both points offer less utility than point X. Investor X will select the portfolio represented by point X.

- The indifference curves illustrated for investor Y are Y_1, Y_2, and Y_3. Investor Y is more risk averse than investor X. This is evident from the fact that the utility curves of investor Y are steeper, with a greater return required for accepting each unit of risk than X. As a result, a portfolio will be selected by Y, which has less risk than that selected by X. It will, however, also have a lower expected return.

2 MULTIPLE-SHARE PORTFOLIO RISK AND RETURN

The principles relating to the calculation of the expected return and the risk of a portfolio with more than two shares are identical to those of a two-share portfolio as per Formula 4.4. The expected return is the weighted average of the expected returns of all the shares in the portfolio. The relevant formula is thus Formula 4.1. Page 4-14

The risk of a multiple-share portfolio, although identical in principle to that of a two-share portfolio, is considerably more onerous to calculate. If, for example, 40 shares are held, the variance matrix will have 1 600 blocks. For each share held, its variance (weighted by the square of the proportion invested in that share), plus its covariance with every other share in the portfolio (weighted by the product of the proportions invested) has to be calculated. For a portfolio of 40 shares, 40 variances plus 780 covariances (each appearing twice) would have to be calculated.

More formally, a two asset portfolio comprises of a 2 × 2 matrix. Similarly a three asset portfolio comprises of a 3 × 3 matrix. It follows that a forty asset portfolio comprises of a 40 × 40 matrix. This matrix contains n terms. In the above example, n represents the individual asset variances. The remaining $(n^2 - n)$ terms comprises the covariances between portfolio assets.

$$(n^2 - n) = (40^2 - 40) = 1\,560$$

or

1560/2 = 780, as each covariance appears twice.

From the formula which follows, it can be seen that the first term is the sum of all the variances, and the second term is the sum of all the covariances.

$$\sigma_p^2 = \sum_{i=1}^{n} W_i^2 \sigma_i^2 + 2 \sum_{i=1}^{n} \sum_{\substack{j=1 \\ i=j}}^{n} W_i W_j \text{cov}_{ij} \qquad \textit{(Formula 4.6)}$$

where: σ_p^2 = the portfolio variance

σ_i^2 = the variance of share i

W_i and W_j = the proportions invested in share *i* and share *j* respectively

cov_{ij} = the covariance between the returns of shares *i* and *j*

The opportunity set of all portfolios of shares on a market such as the JSE can now be graphed as in Figure 4.8.

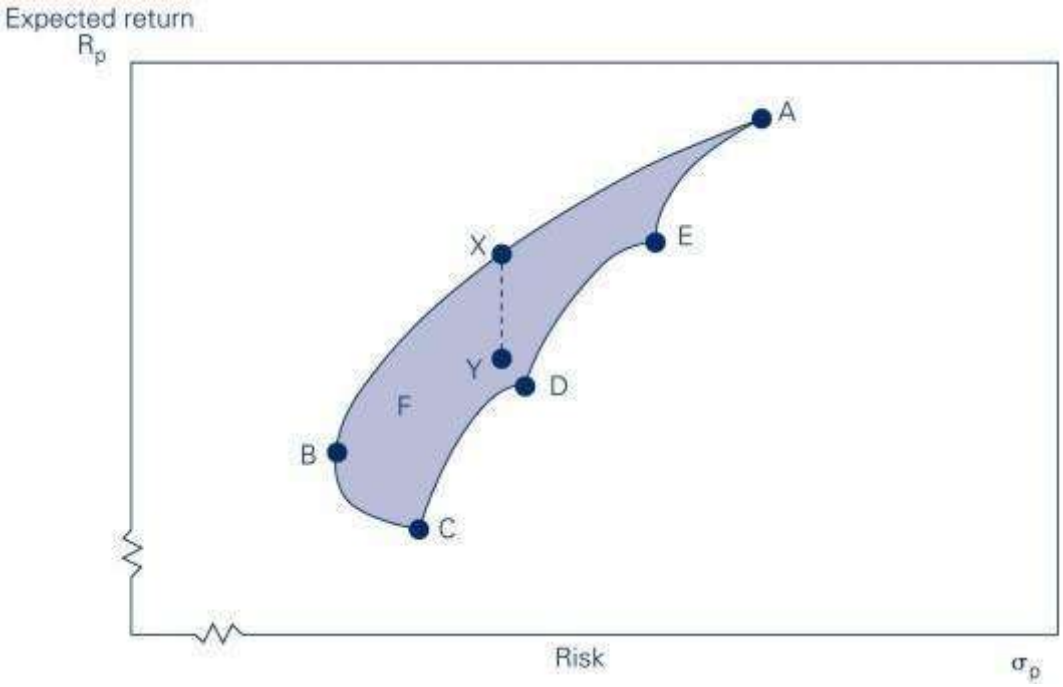


Figure 4.8 Opportunity set of share portfolios

The shaded area represents portfolios of shares available. Page 4-15

The following should be noted from Figure 4.8:

- A, C, D and E are individual shares, which are portfolios of one share.
- The scalloped line AEDC results from the portfolio effect from two-share portfolios of AE, ED and DC respectively.

- All possible portfolios, in the feasible set F (the shaded area), are dominated by the portfolios that lie along AB, the *efficient frontier*. This means that, for any portfolio within the feasible set F, the portfolios along the line AB will offer a superior return to those vertically below the line, for the same risk. (For example, portfolio X offers a higher return than portfolio Y, for the same risk.)
- Investors will invest only in portfolios that lie along the efficient frontier. The portfolio which they will select (that is, the specific point along line AB) will be a function of their risk preference as determined by their indifference curves.

The benefits of diversification

It is evident from the preceding discussion that investors can benefit from allocating their funds to more than a single investment. This process of *diversification* has the effect of reducing the variability of the portfolio returns. Stated differently, the expected return of a portfolio will be the weighted average of returns, but the risk will be less than the weighted average of the variances. In general, it has been shown from research studies that a relatively small number of shares are required in a portfolio in order to derive the major benefit of reduced risk. This is illustrated in Figure 4.9, which also introduces some important concepts of risk. These form the basis of the remainder of this chapter.

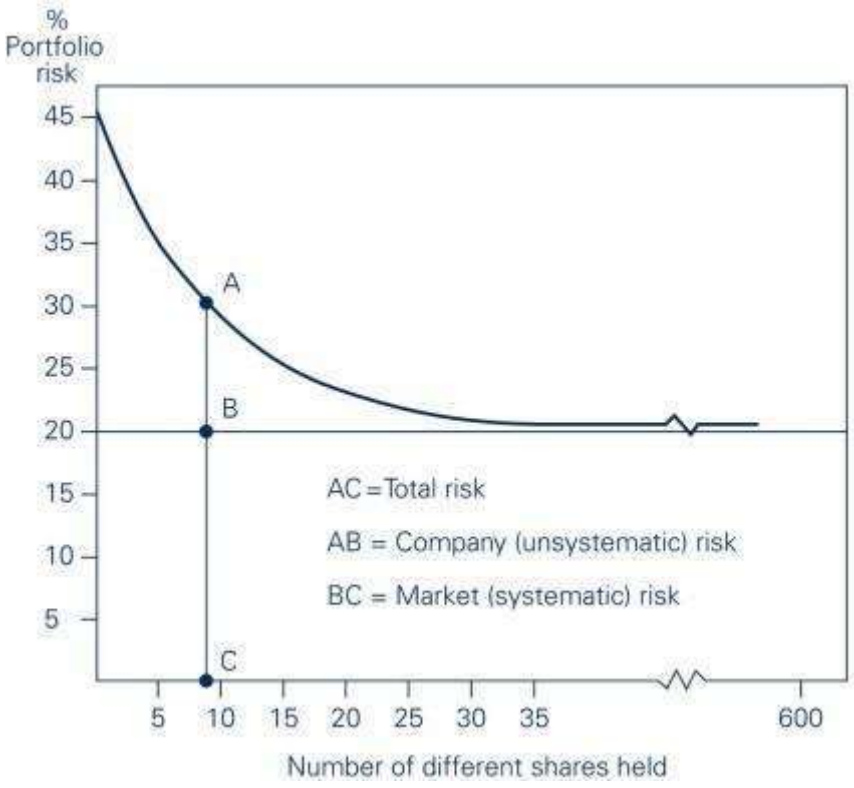


Figure 4.9 Effect of diversification – portfolio risk

The following can be seen from Figure 4.9:

- The risk of the portfolio decreases rapidly with the first few shares held and then levels off until there is no meaningful reduction in risk resulting from an increase in the number of different shares in the portfolio.
- Using eight shares as an example, and reading off line ABC, the total risk of this portfolio is in the region of 30%. Page 4-16
- By enlarging the portfolio to, say, 20 shares, further benefit can be derived in risk reduction, whereafter the marginal benefits of further diversification become considerably reduced.
- A point is reached where all the shares in the market could hypothetically be held. At this point the investor is still exposed to some risk. A market-portfolio risk of 20% is assumed in this example.
- It is useful to distinguish between the two kinds of risk. *Specific risk* relates to variability in returns caused by factors unique to the company, such as the type of industry in which it operates and the product it sells. This is often referred to as *unsystematic diversifiable risk*, or *company risk*. An investor may eliminate this type of risk by diversification.
- The remaining risk results from the vagaries of market sentiment and of economic cycles. It is influenced by factors such as inflation, varying interest rates and foreign exchange rates. All companies, although not equally sensitive to these economic fluctuations, nevertheless tend to be affected in a similar way. This type of risk is generally referred to as *systematic* risk, *non-diversifiable* risk or *market* risk, and cannot be eliminated by diversification.

Introducing a risk-free asset

Assumptions have been made in presenting the case for diversification, and these assumptions underlie a model developed from the work of Markowitz (1959) and Sharpe (1964), generally referred to as *modern portfolio theory* (MPT).

Like most models, the assumptions do not perfectly reflect reality. However, the model can be tested by relaxing the assumptions and determining whether the results obtained are similar to those obtained when applying the assumptions. If this is seen to

be the case, the model is considered to be robust, and to be a reasonable representation of reality. The MPT model discussed so far is based on the following significant assumptions:

- All investors are rational and prefer less risk rather than more for a given rate of return.
- All investors have full and equal access to all available information which results in them all having similar expectations.
- There are no transaction costs such as brokerage; the markets are perfectly competitive; and all financial assets are divisible.
- There is no taxation.

Sharpe (1964) introduced the concept of a risk-free asset together with the assumption that investors can borrow or lend at the risk-free rate. By introducing the risk-free asset, a new dimension of thought about the capital market was made possible. The effect of introducing a risk-free asset such as a treasury bill is illustrated in Figure 4.10.

The following should be noted in Figure 4.10:

- R_f is the risk-free rate at which investors can borrow or lend.
- R_f MC is known as the *capital market line* (CML). It is a tangent to the efficient frontier EF, with M being the market portfolio – i.e. it contains all the financial assets available in the market, proportional to the value of the individual's portfolio.
- The point L_1 represents the portfolio that could be held by an investor L if there were no riskless assets available, and represents a rational choice. However, with the introduction of a risk-free asset, it is clear that investor L could include a proportion of the risk-free asset into the portfolio and move vertically up to point P, thereby enjoying greater expected return for the identical risk at point L_1. However, investor L will act in accordance with his or her particular utility function, thus selecting point L2 rather than point P.

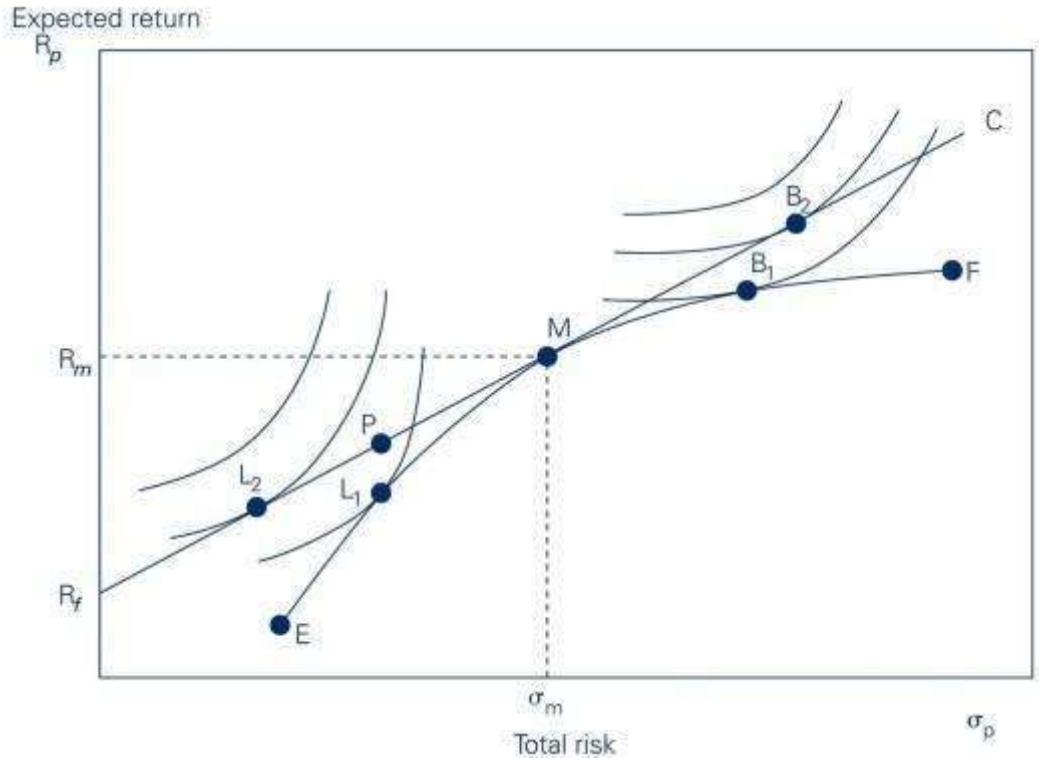


Figure 4.10 The capital market line

- The point L2 represents a point to which investor L could move, based on the indifference curves. At this point, investor L will invest partly in the market portfolio M and will invest (lend) the remainder at the risk-free rate. By so doing, higher utility is achieved as a result of being exposed to lower risk than would have existed had portfolio L_1 been purchased.
- It is of fundamental importance to note that investor L is investing in a combination of the market portfolio M and the risk-free asset R. The proportion of each is dependent upon the point of tangency with line R_fMC. The closer to point R_f the greater the proportion invested in the risk-free asset.
- Similarly, the point B_1 represents the portfolio a less risk-averse investor B would acquire if there were no risk-free asset available. However, with the introduction of the risk-free asset, B will borrow at the risk-free rate and invest personal funds as well as the borrowed funds into the market portfolio M. This results in a move to a higher indifference curve at point B_2. B has thus levered a portfolio by borrowing at the riskfree rate with an obligation to pay interest on the borrowed funds, but investing them together with personal funds in the market portfolio, thus increasing the net expected return.

Example 4.4: Including a risk-free asset in the portfolio

An investor has indifference curves so that her portfolio has R_p = 25% and σ_p = 17% before there is a risk-free asset available on the market. The return on the market R_m = 22%, and the standard deviation of the market, σ_m = 12%. A riskless asset becomes available so that R_f = 12%, and she is able to borrow or lend at that rate. She has an amount of R10 000 available to invest.

This investor is less risk-averse than the average investor (will accept more risk). She can benefit from borrowing and levering her portfolio. If she decides to borrow an additional Page 4-18 R5 000, her new expected return as a result of borrowing may be calculated using the following general formula:

$$R_{pl} = WR_f + (1 - W)R_m \qquad \textit{(Formula 4.7)}$$

where: R_{pl} = the expected return on a leveraged portfolio
W = the percentage of the investor's own funds invested in the risk-free asset (borrowed or lent)

Note that the investor is borrowing in order to invest more money in the market portfolio. In this case, therefore, W is a negative variable as it indicates the payment of interest which will have to be made.

Applying Formula 4.7, the expected return on a leveraged portfolio is:

$$R_{pl} = WR_f + (1 - W)R_m$$
$$R_{pl} = (-0.5 \times 0.12) + (1.5 \times 0.22)$$
$$= 27\%$$

The investor therefore expects to receive a return of 22% on R15 000 and will have to pay interest at 12% on R5 000. A quick check using the actual figures proves this:

Received:	22% × R15 000	= 3 300
Paid:	12% × R5 000	= –600
Return on investment of R10 000		= 2 700

In order to establish the risk of the new portfolio, Formulae 4.6 or 4.5 for risk of a two-asset portfolio are appropriate, the market portfolio being considered as one asset and the riskfree asset as the other. Because the risk-free asset has σ = 0, the formula loses two of its terms and simplifies to:

$$\sigma_{pl} = \sqrt{W_m^2 \sigma_m^2} \qquad \textit{(Formula 4.8)}$$

$$= W_m \sigma_m = 1.5 \times 0.12$$

$$= 18\%$$

The investor, by using the borrowing facility, has levered herself from R = 25% and **σ** = 17% to R***p*** = 27% and σ_{pl} = 18% which offers higher utility to her.

In much the same way, an investor who is more risk-averse than average (prefers less risk), could invest a proportion of available funds into the risk-free asset. The result would be to move down the line R_fMC to a point such as L_2 in Figure 4.11, which would place the investor on a higher utility curve than at point L_1. It is clear from Figure 4.11 that all points along the capital market line in fact dominate the efficient frontier, except point M (the market portfolio, an unlevered portfolio comprising only shares and no riskless asset).

3 BETA ANALYSIS

Modern portfolio theory is one of the most important developments in finance theory. The theory does, however, have practical shortcomings. Firstly, the determination of portfolio risk of a multishare portfolio is extremely onerous. Secondly, the notion that investors would, in practice, purchase a proportion of all the shares in the market is impractical, Page 4-19 although investment in some unit trusts or by buying the all-share index may be comparable to investment in a market portfolio.

It has already been noted that great benefit in risk reduction is achieved by diversifying into a relatively small number of different shares and that the marginal benefit of further risk reduction soon becomes negligible. It has also been noted that company-specific risk, also referred to as *unsystematic risk*, is the element which can be diversified away, whereas investors always remain subject to market risk. Sharpe recognised these shortcomings and devised a measure that reflects the sensitivity of an individual share to fluctuations in the market, thereby measuring the share's non-diversifiable risk. The measure is known as the *beta* of a share.

Because investors will not be rewarded for risk that can be eliminated by diversification, only the market risk as measured by beta is relevant. The measure is called beta because it is represented by the term *b* in the straight line equation $y = a + bx$ – i.e. it is the slope of a line representing the returns on a given share when

compared with the returns of the share market as a whole. This is reflected in Figure 4.11.

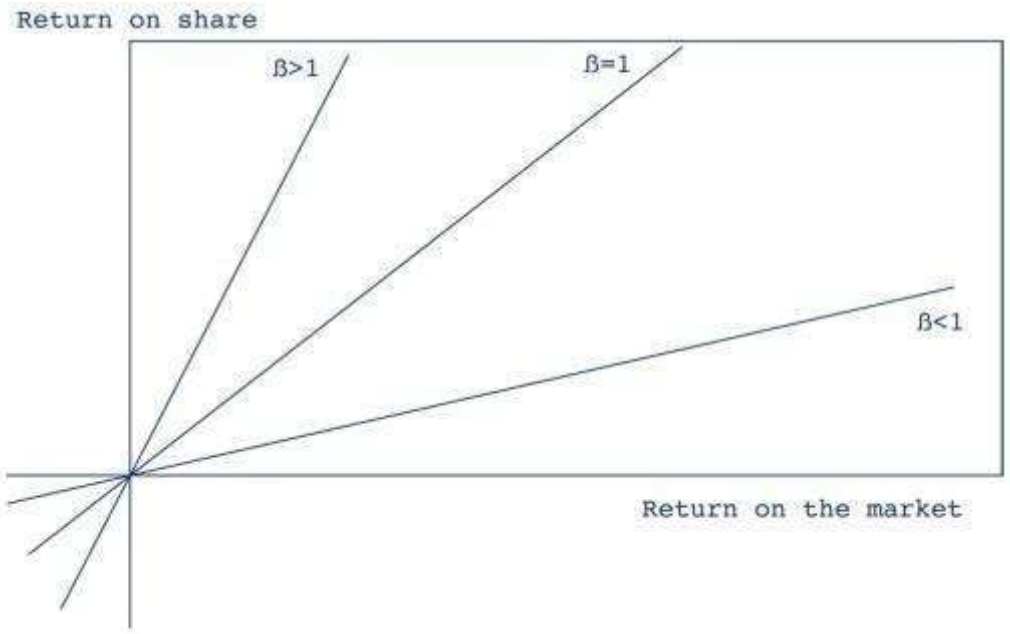


Figure 4.11 Graphic representation of beta

By definition, the market has a beta of one. For example, a share which on average increases its return by 10% when the market returns increase by 10%, and *vice versa*, will also have a beta of one. If a share increases on average by 15% when the market increases by 10%, it has a beta of 1.5. If a share increases on average by 5% when the market increases by 10% it has a beta of 0.5. Beta is thus a measure of the volatility of a share relative to the volatility of the whole share market.

Beta as a measure of portfolio risk

It must be stressed that the future beta of a share is the relevant measure of its market risk. However, the beta is established from past information on the assumption that it will remain fairly stable over time. The total risk of any individual share is:

$$\text{Total risk} = \text{Systematic (market) risk} + \text{Unsystematic (specific) risk}$$

As unsystematic risk can be eliminated by diversification, the risk which is of interest to the investor is the systematic risk as measured by the beta of the portfolio. Page 4-20

The major advantage arising from the use of beta analysis is that the beta of a portfolio of shares is simply the weighted-average of the individual share betas, expressed as follows:

$$\beta_p = \sum_{j=1}^{n} W_j \beta_j \qquad \textit{(Formula 4.9)}$$

where: W_j = the weight of share j in the portfolio

It is now possible to establish the two parameters of risk and return for a portfolio quite easily, provided the betas of the individual shares are known. Investors will therefore manage their portfolios by selecting a portfolio which has a beta that provides them with maximum utility. As beta is a linear concept, it is possible to graph a *security market line* (SML) which is used for displaying the risk-return relationship for individual shares held within a diversified portfolio. The formula for the SML, which is known as the Capital Asset Pricing Model is:

$$R_i = R_f + \beta_i(R_m - R_f) \qquad \textit{(Formula 4.10)}$$

where:
R_i = the required return on share i
R_f = the risk-free rate
β_i = the beta of share i
R_m = the expected return on the market as a whole

It is apparent that, for given shares, the premium for risk ($R_m - R_f$) will be greater as beta increases because that share is more volatile (risky) than the market average. The SML is illustrated in Figure 4.12.

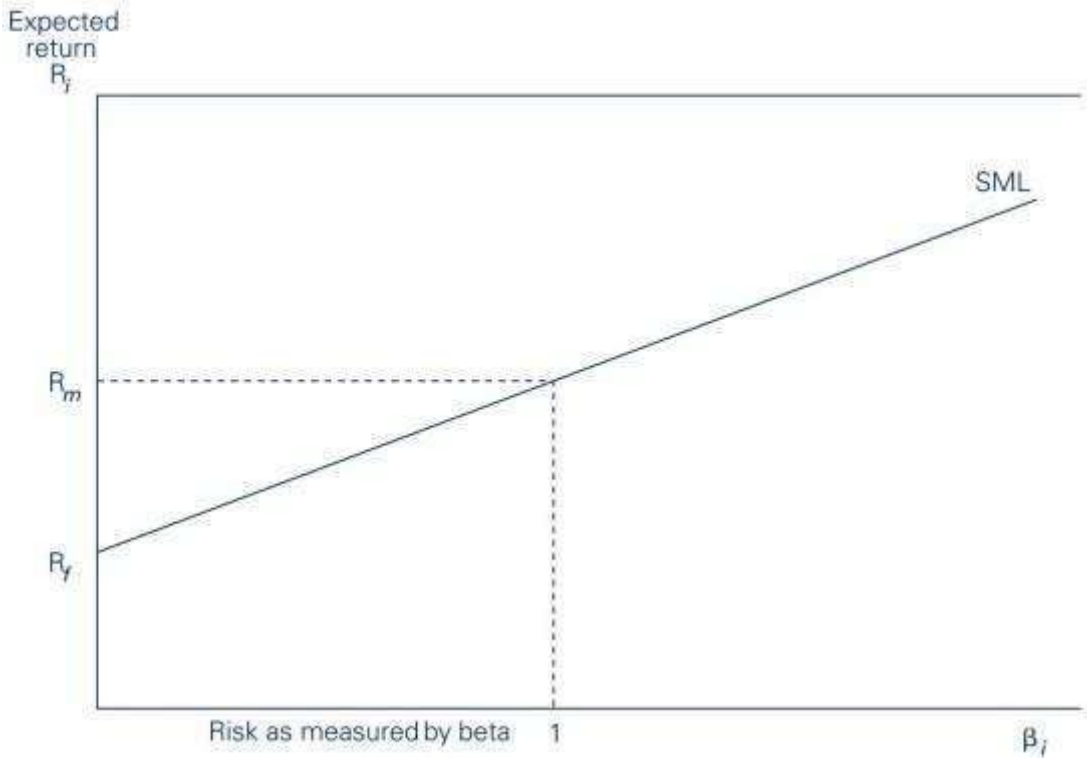


Figure 4.12 The security market line

The expected return of all shares will fall along the SML. This pricing model, known as the *capital asset pricing model* (CAPM), makes it possible for investors to create and adjust portfolios of their choice. Any individual share that does not plot along the SML is either overpriced or underpriced because its expected return differs from its required return, thus requiring a price adjustment to restore equilibrium. If the market is functioning efficiently, the forces of supply and demand will restore the price to equilibrium, on the SML. Page 4-21

Example 4.5: Portfolio management

An investor holds a portfolio of 20 shares with an expected return of 28% and a beta of 1.3. He wants to withdraw 25% of the funds currently invested in share A, which has $R_a = 24\%$ and $\beta_a = 0.9$, and invest them into share B, which has $R_b = 35\%$ and $\beta_b = 2.0$. What is the expected return and beta of the new portfolio?

The expected return of the remaining 75% of the portfolio must first be determined as follows:

$$28\% = (0.25 \times 24\%) + (0.75 \times R_x)$$

The expected return on the remaining 75% of the portfolio R_x is calculated as follows:

$$R_x = [28\% - (0.25 \times 24\%)] / 0.75 = 29.3\%$$

The expected return of the new portfolio will be:

$$R_{pn} = (0.75 \times 29.3\%) + (0.25 \times 35\%)$$
$$= 30.725\%$$

Similarly the beta of the remaining 75% of the portfolio must first be determined as follows:

$$1.3 = (0.25 \times 0.9) + (0.75 \times \beta_x)$$

The beta of the remaining 75% of the portfolio is β_x and is calculated as follows:

$$\beta_x = [1.3 - (0.25 \times 0.9)] / 0.75 = 1.43$$

The beta of the new portfolio will be:

$$\beta_{pn} = (0.75 \times 1.43) + (0.25 \times 2)$$
$$= 1.5725$$

Beta and the capital asset pricing model (CAPM)

In statistical terms, beta is a regression co-efficient expressing the relationship between an individual share and the market with the variance of the market, and it can be determined through least squares regression analysis. Usually weekly or monthly returns are used over a period of years for the company's shares. For a measure of the market as a whole, the market index is used. The procedures for calculating beta are demonstrated in Appendix 4.1. In Appendix 4.2, Obeid Mahomed offers us his perspectives on the determination of betas for the AIFMRM Equity Risk Service (previously the BNP Paribas/Cadiz Equity Risk Service), which is an important provider of betas to practitioners. Betas for South African companies may also be sourced from other services such as Bloomberg and Reuters as well as internally generated betas. We have set out JSE industry sector betas sourced from both AIFMRM and Bloomberg in Table 4.10. The Bloomberg betas and the AIFMRM betas differ as AIFMRM's betas are based on monthly data over 5 years whilst Bloomberg betas are based on weekly data over 2 years. AIFMRM also adjusts for thin trading. It is important to apply judgement in selecting a beta service. Different researchers may arrive at different betas depending on the method of calculation employed, the number of past observations, and other assumptions that have to be made. Betas may be established for individual shares as well as for industry sectors. Betas are subject to fluctuation, although portfolio betas have been found to be fairly stable over time.

Table 4.10 JSE Beta analysis

JSE Sector	Service Provider	Sector Beta	Selected large companies in sector
Telecom Service Providers	AIFMRM	0.80	MTN, Vodacom
	Bloomberg	0.57	
Personal Care, Drug & Grocery	AIFMRM	0.53	Pick n Pay, Shoprite, Clicks,
	Bloomberg	0.60	
Retailers	AIFMRM	0.95	Woolworths, Pepkor, Mr Price, Foschini, Truworths
	Bloomberg	0.73	
Personal Goods	AIFMRM	1.19	C F Richemont
	Bloomberg	1.36	
Pharmaceuticals & Biotechnology	AIFMRM	0.94	Aspen, Adcock Ingram
	Bloomberg	0.49	
Travel & Leisure	AIFMRM	1.19	Tsogo Sun, Sun International, Famous Brands
	Bloomberg	0.53	
Industrial Transport	AIFMRM	0.99	Textainer, SuperGroup
	Bloomberg	0.57	
Industrial Support Services	AIFMRM	0.62	Hudaco, Mondi
	Bloomberg	0.10	
Food Producers	AIFMRM	0.32	AVI, Astral, Tiger Brands, RCL Foods, Oceana
	Bloomberg	0.37	
Beverages	AIFMRM	0.70	Anheuser-Busch Inbev
	Bloomberg	0.21	
Construction and Materials	AIFMRM	0.78	WBHO, PPC
	Bloomberg	0.62	
Consumer Services	AIFMRM	0.89	Advtech, Curro
	Bloomberg	0.51	
Healthcare Providers	AIFMRM	0.50	Netcare, Life Healthcare, Afrocentric
	Bloomberg	0.35	
Software & Computer Services	AIFMRM	0.62	Prosus, Naspers N, Bytes Tech, Datatec, Karooooo
	Bloomberg	1.37	
Electronic and Electrical Equipment	AIFMRM	0.60	Reunert
	Bloomberg	0.73	
Precious Metals and Mining	AIFMRM	1.26	Amplats, Gold Fields, AngloGold, Harmony
	Bloomberg	1.53	
General Industrials	AIFMRM	0.88	Bidvest, Barloworld
	Bloomberg	0.58	
Chemicals	AIFMRM	1.70	Sasol, AECI, Omnia
	Bloomberg	0.95	
Non-Life Insurance	AIFMRM	1.70	Santam, Outsure
	Bloomberg	0.43	
Life Insurance	AIFMRM	1.02	Discovery, Old Mutual, Momentum Met., Sanlam
	Bloomberg	0.66	
Industrial Materials	AIFMRM	0.80	Sappi
	Bloomberg	0.78	
Industrial Metals & Mining	AIFMRM	1.20	ARM, BHP, AngloAmerican, Glencore, South32, Kumba
	Bloomberg	1.38	
Banks	AIFMRM	0.94	FirstRand, Capitec, ABSA, Standard Bank
	Bloomberg	0.69	
Real Estate Inv. Trusts	AIFMRM	1.03	Growthpoint, Vukile, Hyprop, Redefine, Hammerson, Emira
	Bloomberg	0.54	

Note: AIFMRM betas are based on monthly data for the previous 5 years, and subject to Bayesian adjustments and thin-trading adjustments whilst Bloomberg is based on weekly data for the previous 2 years. You can also obtain individual company betas from Reuters, ShareData Online (click on Shares and on the company) and Yahoo Finance for the larger companies.

Table 4.10 illustrates betas calculated for sectors on the JSE. A Bloomberg beta of 1.53 for the Precious Metals and Mining sector indicates that this sector is relatively more volatile than the overall index. This means that if the overall market changes by 1%, then the Precious Metals and Mining sector changes in the same direction by 1.53%. Betas for some sectors are larger than others.

Food Producers are on the other side of the beta spectrum, with an industry beta of 0.37. This is significantly lower than the market beta of one and reflects the lower risk of investing in this sector. Food Producers are expected to offer stable cash flows. Some other betas seem to offer questionable results and we need to apply betas with a degree of judgement and understanding of the underlying statistical dimensions of the data employed. In South Africa, the resources sector constitutes about 20-30% of the market capitalisation of the JSE. This means that the betas of firms outside the resources sector will be lower in relation to the All Share Index because of the resources sector and its higher volatility.

Returns are measured as a percentage increase or decrease in the share price (plus dividends received) at the end of the month, if monthly readings are taken, over the share price at the end of the previous month. The equation for finding the beta of a company in simple regression form if monthly data are used is:

$$R_i = \alpha + \beta_i(R_m) + e \qquad \textit{(Formula 4.11)}$$

where: R_i = the monthly return on the share i
α = the intercept on the vertical axis
β_i = the slope of the regression line for share i
R_m = the monthly return on the market
e = a random error term

In order to find β, the usual statistical procedures for regression are required, solving for α (alpha) and β (beta). It should be noted that the random error term *e* is expected to be zero Page 4-23 as random errors should sum to zero. The intercept alpha should also be zero when excess returns are used. Any deviation from zero in the alpha indicates that the share has at some time been inefficiently priced, a situation which is unlikely to persist if the market is efficient.

Figure 4.13 illustrates the process used to establish the beta of share *i*. Monthly readings have been taken of the return on the market as measured by the change in the market index and plotted against the returns on share *i*, adapted for monthly returns. It is clear that share *i* has a low beta, probably in the region of 0.5. Its low beta results from the fact that it is clearly less volatile than the market as a whole. Investors would therefore expect to receive a return less than that of the market. The line has an alpha which is greater than zero in this instance, indicating a price appreciation over this period which was in excess of that of the market as a whole. In general, this relative appreciation will not be expected to continue indefinitely.

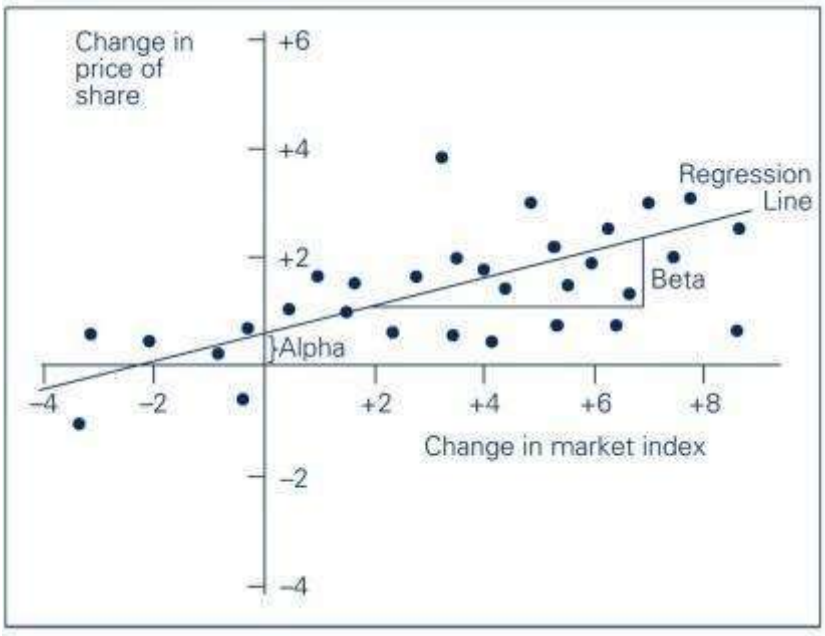


Figure 4.13 Regression analysis to establish market risk (beta)

Once the beta of an individual share is known, it is relatively easy to apply the CAPM since the risk-free rate and expected return on the market are usually readily available.

Example 4.6: Capital asset pricing model

An investor wants to add shares in Investgro Ltd to her portfolio. Investgro has a beta of 0.5, the risk-free rate is 7%, and the return on the market is 12%. Establish the

expected return on Investgro.

Using Formula 4.10:

$$R_i = R_f + \beta_i (R_m - R_f)$$
$$= 0.07 + 0.50(0.12 - 0.07)$$
$$= 9.5\%$$

The expected return on Investgro Ltd, as anticipated, is less than the 12% expected return on the market. The investor would now attempt to establish whether the actual return is likely to be higher or lower than 9.5%. If it is likely to be higher, the share is underpriced and should be bought. Using the SML, the relationship can be plotted and Investgro's position established as in Figure 4.14. Page 4-24

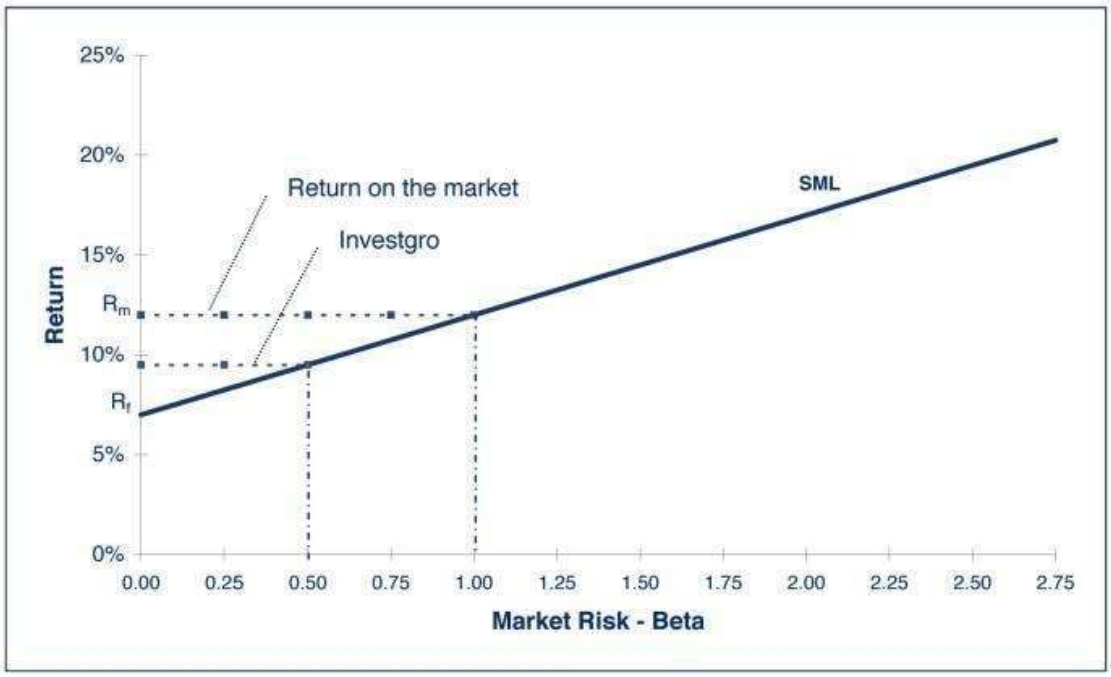


Figure 4.14 SML to plot share *i*

CAPM is an elegant theory which is based on some restrictive assumptions. As we relax some of these assumptions we come into contact with reality. The CAPM offers a valuable framework for analysing risk and return but we need to reflect on its practical usefulness by evaluating the assumption of perfect capital markets or efficient markets.

4 THE EFFICIENT MARKETS HYPOTHESIS

Perhaps the most significant assumption of modern portfolio theory and the CAPM is the assumption of perfect capital markets. In a less rigorous form, the assumption is that all capital markets are efficient. The efficient market hypothesis (EMH) posits that a company's share price reflects all available information. Fama (1970) suggested three forms of market efficiency.

The weak form

The first level of efficiency, known as the *weak form* of the EMH, holds that share-market prices follow a *random walk*. This is likened to the path of a drunk where it is impossible to predict whether the next step will be to the left or right. The information impounded in the share prices is thus considered to fully reflect the historic price sequence of each individual share. As a result, it follows that price changes are independent of one another, making it impossible to predict a future price based on a series of past prices. In terms of the weak form of the EMH it should not be possible to employ trends, charting, technical analysis and past movements in share prices to earn superior returns.

The semi-strong form

The *semi-strong form* of the EMH holds that all publicly available information about a share is impounded immediately and without bias into the share price. The historic price sequence is also included in the information set. It is not possible to use fundamental analysis, such as the analysis of financial statements, to earn superior returns. This is because such information has already been absorbed by the market and is immediately reflected in a price change. The share price reflects all publicly available information. Only with inside information is it possible to identify shares that are incorrectly priced.

An interesting aspect of efficiency is that it requires the presence of investors who do not accept the efficiency of the market. As a result, they are engaging in fundamental analysis in an attempt to discover new information. This activity in turn contributes to market efficiency.

The strong form

The third level of efficiency, known as the *strong form*, holds that all information, both privately and publicly held, is impounded into the share price immediately and without bias. Thus it is impossible for any investor to consistently outperform the market, even with "inside" information.

Market efficiency: the evidence

The evidence about market efficiency is mixed and complex. Some of the evidence that supports market efficiency includes the following findings:

- Although there is some relation between movements in share prices over short horizons, these are not sufficient to cover trading costs. This supports the weak form of market efficiency.
- Share prices react quickly and in the right direction to new information that is price sensitive such as mergers and acquisitions, earnings and dividend announcements.
- Professional fund managers on average have not been able to beat the market index after fees.

What is the evidence that indicates that markets are not efficient? This is where it gets hard, for either markets are inefficient or, alternatively, CAPM does not capture all the relevant risk factors. We will present these for now as market anomalies:

- When we analyse price movements over 3- to 9-month holding periods, we find a **momentum** effect as companies that have done well will continue to do well and companies that have done poorly will continue to do poorly in later periods – over a subsequent period of about 3 years. Investing in a portfolio of the best performing shares over the last six months and undertaking short sales of the worst performing shares over the previous six months will result in positive returns relative to other shares. Therefore, there seems to be evidence of a momentum effect over periods of about three years. The indication here is to invest in past winners and short past losers.
- Over longer multi-year periods, we find that there may be negative long-term performance effects. This means that companies that have performed poorly over long periods of 3 or 4 years will see a reversal of fortune and will do well in a subsequent period whilst companies that have performed strongly over a period of three or four years will perform poorly in the years thereafter. This effect is also called **mean reversion**. This leads to a contrarian investment philosophy whereby it pays to buy the long-term losers and sell the long-term winners. This may indicate that there is overreaction over intermediate horizons, leading to the

momentum effect, which is then followed by a long-term reversal as investors come to terms with past overreactions to positive and negative news.

- A portfolio of **low price–earnings (P/E) shares** have generated higher returns than a portfolio of high P/E ratio shares. This could be because low P/E ratio companies involve higher risks not captured by CAPM betas.
- **Neglected firms** are firms not covered by the large institutional investors nor researched by brokers and analysts. Neglected firms and **small firms** tend to earn higher returns than that predicted by CAPM. This again may indicate that the risks of investing in small firms and neglected firms, where there is little research or information or liquidity, will increase investment risk which means a higher return is justified.
- An investment strategy of investing in firms with **high book-to-market value of equity** ratios will result in higher returns.
- Although share prices react quickly to earnings announcements, companies that report positive earnings experience a **subsequent upward drift** in share returns whilst negative earnings announcements may result in a downward drift in subsequent returns.

Page 4-26 It would be naïve to expect that **inside information** will not lead to abnormal profits. This is why there should be and there are penalties for trading on the basis of inside information.

Value shares, which are shares with high book-to-market ratios, low P/E ratios and depressed share prices, were found to have offered higher returns than high-growth shares. Size effects, momentum and high book-to-market ratios have been found to apply in securities exchanges across many countries. Fama & French indicate that these factors are proxies for risk and they have added book-to-market and size factors to capture the required return. This extends CAPM by adding these two factors to the model and is referred to as the Fama-French Three-Factor Model. We set this out in Chapter 7.

It is important to note that some of these anomalies have since gone missing. Some anomalies do not seem to stand the test of time and the publication of such anomalies can have an effect on the effectiveness of these anomalies thereafter. To conclude, we would state that the evidence for market efficiency is mixed and we will set out how firms adjust the CAPM cost of equity in Chapter 7.

Evidence of the efficiency of the JSE

The weak form of the EMH was found to apply on the JSE on the basis of analysing short-term movements in share prices. Van Rensburg and Robertson[2] have indicated that the small size effect holds for the JSE and investing in low price–earnings companies will offer higher returns than indicated by CAPM. Applying the Fama-French Three-Factor Model seems to explain the returns of companies on the JSE. McKane and Britten[3] reported a significant liquidity premium for the JSE in that investors require a higher return for holding illiquid shares. The JSE is significantly less liquid than the major securities exchanges and there are closely held and thinly traded shares. Even relatively large and well-known companies may not trade each day of the year and we have set out some of these companies with their percentage of days traded. This does not even consider the impact of volumes on pricing and only reflects whether there was trading in the shares each day of the year. Investors require higher long-term returns for holding relatively illiquid shares. The liquidity and tradability of small, listed companies is significantly worse than the following companies which are relatively large companies and yet only trade between 49% to 99% of the time.

Company	% days traded	Company	% days traded	Company	% days traded
Brimstone N	72.0%	Italtile	99.0%	Spur Corp.	98.0%
NuWorld	75.0%	York Timber	87.0%	SASFIN	76.0%
Crookes	49.0%	CMH	97.0%	Bell Equip	86.0%
KAL (Kaap Agri)	94.0%	Choppies	56.0%	Caxton	91.0%

[Source: AIFMRM Equity Risk Service – June 2023]

How have professional fund managers performed in South Africa? In a study by Wessels and Krige,[4] it was found that although fund managers were able to achieve short-term quarter- to-quarter persistence in returns, the longer-term persistence in performance tended to disappear. Bradfield and Swartz[5] found that the top quartile funds indicated a high degree of persistence of returns from quarter to quarter. However, persistence declined over longer Page 4-27 periods. However, there were a few funds that exhibited persistence in either outperforming or underperforming the ALSI.

Let's evaluate for example the investment performance of Allan Gray. In the 49 years to 30 June 2023, Allan Gray Equity has managed to achieve an annualised return of 23.6% as compared to the annualised return of 17% for the ALSI. This has been achieved at a slightly lower level of risk. What does a 6.6% per year outperformance

by Allan Gray mean over 49 years? If you invested R1000 in 1974 in the JSE ALSI, it would have accumulated to R2.2m by June 2023 whilst R1000 invested in Allan Gray would have accumulated R32.4m! The evidence is that most managers are unable to beat the market index, but there are a few exceptions. However, we need to understand that Allan Gray is active sometimes in affecting managerial performance and decisions.[6]

What about passive investing in South Africa? For example, investors are able to invest in the Satrix 40 index fund which is offered by the JSE at a very low cost, and passive index fund investing is expected to grow in South Africa.

Is the CAPM used in practice?

Yes. In a PWC survey of South African investment professionals, it was found that the CAPM is by far the most dominant method used in practice to determine a company's cost of equity. The CAPM is based on the use of betas, and Bloomberg and the AIFMRM Equity Risk Service are popular sources of betas. Reuters, FactSet, ShareData, Yahoo Finance and CapitalIQ also provide company betas. The use of betas in determining the cost of equity is dominant in South Africa and in many other countries such as the USA, the UK and Australia. Yet, Warren Buffett thinks differently. As he states in one of his letters to his shareholders:

> To invest successfully, you need not understand beta, efficient markets, modern portfolio theory, option pricing or emerging markets.

What do we think? The use of betas makes sense but we may need to apply judgement in particular cases by analysing the underlying data and we need to take care not to double count for risk. Further, there are other issues in determining the cost of equity and we will come back to this when we study the cost of capital in Chapter 7.

Does CAPM work? The evidence from empirical studies is that we need to extend CAPM in order to explain share returns adequately. The SML is flatter than indicated by CAPM. Fama- French (1992) was able to increase the accuracy of predicting share returns significantly by including size and value factors, as well as a market beta factor. Small companies and companies with high book values relative to their share price (value shares) tend to produce higher returns than indicated by beta alone and according to Fama-French, these represent risk factors. There are therefore three sources of risk – market risk, size risk (investing in smaller companies implies higher risk) and high book-to-market ratios (which will often include companies that are

experiencing financial distress or changes to their business models). The **Fama-French Three-Factor Model** includes a market beta factor, as in CAPM, and then includes a size factor and a value factor to predict a company's expected/required return. In the annexure to Chapter 7, we explain this model in greater detail and we also indicate how practitioners make ad-hoc adjustments to CAPM to take into account these factors in determining a required return. Page 4-28

5 EXCHANGE TRADED FUNDS (ETFs) AND UNIT TRUSTS

In order to acquire an investment that replicates the performance of the JSE All Share or the JSE Top 40 index, an investor would need to buy all the shares making up the JSE All Share or the JSE Top 40 index. A small investor may not have sufficient funds to achieve this and transaction costs would be high. Yet, an investor may wish to invest in a portfolio of shares to attain diversification.

In order to achieve effective diversification, the investor may invest and acquire units in a unit trust such as the Allan Gray Equity Fund. You send your money to Allan Gray and this is combined with the contributions from other investors to acquire shares in companies. You do this by purchasing units in a unit trust at its net asset value which reflects the values of the underlying equity investments owned by the unit trust at the end of each day. As the unit trust receives funds from investors, it purchases shares in line with its investment policy. An index fund which tracks the JSE Top 40 would purchase shares in the JSE Top 40 companies in line with their market value weightings in the index. If many investors wish to withdraw funds then the unit trust will need to sell shares in order to meet its redemption obligations. A fund will retain a small percentage of its funds in cash in order to meet liquidity requirements and meet redemptions but when markets fall and many investors withdraw funds from a unit trust at the same time, then the unit trust will need to sell shares which can then further accentuate the selling pressure on shares. This makes a unit trust an open-ended investment entity.

Exchange Traded Funds (ETFs) are similar to unit trusts but are listed on the JSE. Essentially an investor can acquire shares in an ETF which owns a portfolio of shares or bonds or commodities. Investors are able to trade ETFs like any other share on the JSE but the ETF represents ownership of a portfolio of assets which may offer an investor the benefits of diversification at a low cost. A commodity ETF will invest in gold bullion, or platinum or another commodity. If an investor wished to invest in

gold, then often the investor would either need to invest in physical gold which is not practical[7] or more likely acquire shares in gold mining companies such as Gold Fields. However, an investor may not wish to be exposed to the risks of gold mining in regard to changing ore grades, higher operating costs, production disruption due to seismic activity, industrial action and regulatory risks. An investor may simply acquire shares or debentures in a gold ETF which acquires gold bullion, and the value of the gold ETF will reflect the underlying value of its gold holdings as the gold price moves up and down.

Another ETF may simply reflect an index such as the JSE Top 40 index. The advantages of investing in an ETF include diversification, flexibility with regard to trading, low costs, liquidity and transparency. An index ETF simply replicates the underlying index, which requires no active management and therefore costs are low. The Satrix Top 40 ETF has a cost of 0.10% (but there are trading costs) whilst the Satrix Top 40 Unit Trust has a direct cost of 0.55%, including VAT (in 2023). An actively managed equity unit trust fund may involve fees to investors of about 1.0% to 2.2% of the funds under management per year.

The largest gold ETF is the **NewGold** Exchange Traded Fund managed by ABSA's CIB division. ABSA states the following about NewGold:

> Absa CIB's NewGold Exchange Traded Fund (NewGold) is one of the simplest and cost-efficient methods for investors to invest directly in actual gold. NewGold continuously tracks the gold spot price and enables investors to invest in a listed instrument (structured as a debenture) in which each security is equivalent to approximately 1/100th ounces of gold and is fully backed by holdings of gold bullion

The NewGold ETF had a total NAV of over R18bn in 2023 and the ETF holds 457 526oz (12 971 kg) of physical gold bullion. The annualised total expense ratio is 0.3%. Page 4-29

Let's analyse a possible scenario. Thabisa and Zanele are Chartered Accountants and friends who decided in 2008 to invest R1m each in gold. Thabisa decided to invest her R1m in the NewGold ETF. Zanele decided to invest her R1m evenly across three major gold mining companies, Gold Fields, AngloGold and Harmony. Ten years later, in September 2018 and almost 15 years later in June 2023 you plot how they have done with their gold investments. This is presented in Figure 4.15.



Figure 4.15 Investing in Gold: NewGold ETF versus gold mining company shares

Thabisa would in September 2018 have R2.37m whilst Zanele would only have about R373 333 [(0.46+0.40+0.26)/3 x R1m] even though they both started with R1m each. Zanele would have endured the risks of gold mining, whilst Thabisa was only exposed to the risk of the underlying gold price movements[8]. Zanele decided to stay invested in the mining companies. By May 2023, Thabisa's NewGold ETF was now valued at R5.02m whilst Zanele would now have R1.8m [(3.05 + 1.48+0.88)/3 x R1m]. Although the gold mining companies outperformed NewGold in the five years to 2023, the NewGold ETF significantly outperformed the mining company shares over the whole period of 2008 to 2023. This indicates the potential value of ETFs to portfolio management.

Satrix manages over R180bn of index tracking assets as well as smart beta strategies, and provides investors the ability to invest in very low-cost index funds and ETFs. For example, it offers investors access, smart beta strategies to the Satrix 40 fund and Satrix 40 ETF. Satrix states the following: Page 4-30

> The **Satrix 40 fund** tracks the performance of the FTSE/JSE Top 40 Index which represents the 40 largest companies by market cap. It endeavours to accurately replicate the index by holding all constituents in the exact index weighting. The index is rebalanced quarterly and the fund therefore incurs minimal trading fees as a result of low turnover. This fund is ideal for investors who seek general market performance while minimising trading costs.

The **Satrix 40 ETF** tracks the performance of the FTSE/JSE Top 40 index which includes the 40 largest companies on the JSE, ranked by investable market cap. The ETF aims to accurately replicate the index by holding all constituents in the exact index weighting. Distributions are made quarterly.

As ETFs trade as shares, the share price of an ETF can divert away from NAV and sometimes ETFs may trade at a discount to NAV For example, assume that the market value of the XYZ ETF's holdings adds up to R660m and there are 600m shares in issue. We would expect that the value of the ETF share price should be R1.10 (660/600). However, it may trade at R1.04 which is at a discount to NAV but represents the buying and selling actions by investors in the ETF. It does also mean that management is not obligated to dispose of shares or assets owned by the ETF, which may be useful during periods of short-term but severe market declines. Investors redeem their investment by selling their ETF shares on the JSE like any other share. In times of market stress, there is some evidence that the prices of ETFs may be more volatile than the market.

An ETF will involve brokerage costs and there is a spread between buying and selling rates. Liquidity may be an issue although ABSA for example may act as a market maker to ensure liquidity for its ETFs. There are now ETFs for commodities, industry sectors, factor investing strategies, high-dividend yield shares (e.g. SATRIXDIV), low-volatility shares, emerging markets and so on. ETFs enable investors to diversify beyond equities and bonds, but investors should still be cautious with regard to the nature of the underlying assets of any ETF and the reputation of the issuers[9].

FROM THE REAL WORLD

You may ask whether the principles of diversification are relevant in the investment decisions undertaken by asset managers such as Allan Gray, Sygnia and Coronation. The following are extracts from their correspondence with their investors, which indicate the importance of diversification in reducing portfolio risk. Seema Dala of **Allan Gray** writes:

> *"Diversification is only useful when the assets being added to the portfolio have low correlations with the existing assets. As we add asset classes to a portfolio, which have negative or zero correlations to the assets, which are already in the portfolio, the portfolio becomes more diversified. This should in turn reduce the volatility, or unpredictability, of portfolio returns."* With regard to developed markets, she states the following: *"... asset classes are highly uncorrelated to the ALSI, so adding these (developed market) assets to a portfolio of local assets would significantly reduce overall portfolio volatility" (Quarterly Commentary Q1: March 2013)*

The **Sygnia** Skeleton Balanced Absolute Fund in March 2023 states the following in relation to the fund's risk profile:

> *The risk in the fund is managed by spreading investments across asset classes that deliver uncorrelated returns over time, ensuring a diversification of sources of returns over market cycles.*

Coronation writes the following in relation to its Balanced Plus Fund in March 2023:

> *The fund's investment in shares is carefully balanced with other assets (including cash, bonds and property) to ensure that risk is moderated. Returns from these assets are not as volatile as shares and will not always move in the same direction (up or down) at the same time, making the fund less risky than a pure equity fund.*

Page 4-31 SUMMARY

The rationale for investing in a portfolio of shares rather than a single share is based on the benefit received in the form of *risk reduction* by diversification. The expected return of a portfolio is the weighted average return of the shares held in the portfolio. The risk, however, is not a weighted average of risks because of the effect of less than perfect correlation of returns among the shares in the portfolio. *The capital asset pricing model* may be an appropriate method of pricing individual shares held by a diversified investor. Establishing the *beta* of a company enables the effect of that particular share on the overall portfolio to be assessed. Using beta analysis it is possible to manage a portfolio of shares in accordance with the desired level of market risk, the specific risk having been eliminated through diversification.

Market efficiency is a fundamental assumption which has significant implications for financial management.

GUIDANCE TO PORTFOLIO MANAGEMENT

By Johnathan Dillon M.Com CA(SA)

Summary of the risk and return concepts linking Chapters 3 and 4

Chapter 3 deals with the concepts of risk and return in relation to a single share, as well as with the comparison of single shares. Chapter 4 does the same, except in relation to a portfolio of shares. The table below presents a summary of the pertinent concepts of these chapters relevant to analysing shares:

	RETURN	RISK	COMPARISON
1 share	Expected return (R_a) of share	Standard deviation (σ) or variance (σ^2) of share ***Or*** Beta (β) of share	Co-efficient of variation (CV) ***Or*** *z* score
2 shares (portfolio)	Weighted average of shares' expected returns (R_p)	Standard deviation (σ) or variance (σ^2) of the portfolio, calculated as: $\sigma_p = \sqrt{\sigma_p^2}$ $\sigma_p^2 = W_a^2\sigma_a^2 + W_b^2\sigma_b^2 + 2W_aW_b\text{cov}_{ab}$ ***Or*** Weighted average of shares' betas (β)	Co-efficient of variation (CV) ***Or*** *z* score
3+ shares (portfolio)	Weighted average of shares' expected returns (R_p)	The standard deviation (σ) or variance (σ^2) of the portfolio can be determined using a lengthy formula similar to that used under 2 shares above ***Or*** Weighted average of shares' betas (β_p)	Co-efficient of variation (CV) ***Or*** *z* score

Page 4-32 A summary of important risk symbols is presented in the table below:

SYMBOL	NAME OF SYMBOL	SUMMARY OF WHAT IT MEASURES
σ_a^2	Variance of share *a*	The extent to which the expected returns of the share deviate from the mean.
σ_a	Standard deviation of share *a*	The same as what the variance measures – it is merely the square root of the variance and in a standarised form.
CV_a	Co-efficient of variation of share *a*	Relates the units of return to the units of risk by measuring the units of risk per 1% of return.
Cov_{ab}	Co-variance of two shares *a* and *b*	The extent to which the returns of two shares move together and in which direction (that is, the relationship between the returns of the two shares).
ρ_{ab} OR r_{ab}	Correlation co-efficient of two shares *a* and *b*	The same as what the co-variance measures – it is merely the standardised form of the co-variance as it falls between –1 and +1.
r_{ab}^2 OR R^2	Co-efficient of determination of two shares *a* and *b* (also known as R-squared)	The proportion of variability (the goodness of fit) between two data sets (in this instance, share *a* returns and share *b* returns) and is merely the square of the correlation co-efficient; hence, it will be between 0 and +1.
β	Beta of a share	The non-diversifiable risk of a share as reflected by the share's sensitivity to fluctuations in the market.

The important risks covered in these chapters, as well as how they link together and are measured, are summarised below:

Total risk for an individual share or portfolio can be expressed as follows:

Total risk = systematic (market) risk + unsystematic (specific) risk

Total risk is measured by calculating the variance or standard deviation of the returns of the share or portfolio. However, when taking into account the concepts of business and financial risk covered in Chapter 3, the total risk of an entity that shareholders are exposed to can also be expressed as follows:

Total risk = business risk + financial risk

A somewhat useful measure of this expression of total risk is the degree of combined leverage (DCL), which equals contribution divided by profit before tax or the degree of operating leverage (DOL) multiplied by the degree of financial leverage (DFL). However, as noted in Chapter 3, the DCL does not entirely measure all facets of business and financial risk. This expression of total risk also links directly with the asset beta and equity beta concepts covered in greater detail in Chapters 7 and 10, as well as with the need to "unlever" and "relever" betas in certain situations. Page 4-33

Systematic risk (also referred to as **non-diversifiable** or **market risk**) is the risk that remains after diversification, that is, the risk of being in the economic system in which the entity operates. No amount of diversification can eliminate this risk. Systematic risk is measured by calculating the beta of a portfolio or share's returns in relation to the returns on a fully diversified market portfolio.

Unsystematic risk (also referred to as **diversifiable**, **specific** or **company risk**) is the risk of a portfolio in excess of the systematic risk. Investors can reduce or even eliminate this risk by holding more and more shares until the portfolio risk equals the market / systematic risk.

Sovereign risk is essentially the risk of a country based on economic, political and financial factors, that is, the country's financial health and creditworthiness, which have a bearing on the systematic risk of a market. Therefore, the higher the sovereign risk of a country, the higher the systematic risk of an economic system and consequently the higher the risk of all entities operating within that particular country.

Business risk is the risk relating to the operating activities of the entity, that is, the risk inherent in its operations ignoring how the entity's operations are financed.

Financial risk is the additional risk placed on an entity (and on ordinary shareholders) as a result of the decision to finance with debt rather than equity.

SELF-STUDY PROBLEM

Shares A and B have the following historical dividend and price data:

	A	B	C	D	E	F
1		Share A			Share B	
2	Year	Dividend	Year-end Price		Dividend	Year-end Price
3	20.2		22.50			43.75
4	20.3	2.00	16.00		3.40	35.50
5	20.4	2.20	17.00		3.65	38.75
6	20.5	2.40	20.25		3.90	51.75
7	20.6	2.60	17.25		4.05	44.50
8	20.7	2.95	18.75		4.25	45.25

Required:

(a) Calculate the realised rate of return (or holding-period return) for each share in each year. Assume an equally weighted portfolio. What would the realised rate of return on the portfolio be in each year from 20.2 through 20.7? What are the average returns for each share and for the portfolio?

(b) Calculate the standard deviation of returns for each share and for the portfolio.

(c) Based on the extent to which the portfolio has a lower risk than the shares held individually, would you assess that the correlation co-efficient between returns on the two shares is closer to 0.9, 0.0 or – 0.9?

(d) If you added more shares at random to the portfolio, what is the most accurate statement of what would happen to σ_p ?

- σ_p would remain constant, or
- σ_p would decline to somewhere in the vicinity of 15%, or
- σ_p would decline to zero if enough shares were included. Page 4-34

Suggested solution

(a) The return to the shareholder for each period for holding a share is calculated as follows:

$$\text{Return} = \frac{\text{Dividends for the period} + \text{Share price at the end of period}}{\text{Share Price at beginning of period}} - 1$$

The returns are presented in the following table:

	A	B	C	D	E	F	G	H	I
10		Share A			Share B				
11	Year	Dividend	Year-end Price		Dividend	Year-end Price	Return $_A$ $[(P_1+D_1)/P_0 - 1]$	Return $_B$ $[(P_1+D_1)/P_0 - 1]$	Return $_{Portfolio}$ $[0.5xR_A+0.5xR_B]$
12	20.2		22.50			43.75			
13	20.3	2.00	16.00		3.40	35.50	**-20.00%**	**-11.09%**	**-15.54%**
14	20.4	2.20	17.00		3.65	38.75	20.00%	19.44%	19.72%
15	20.5	2.40	20.25		3.90	51.75	33.24%	43.61%	38.42%
16	20.6	2.60	17.25		4.05	44.50	-1.98%	-6.18%	-4.08%
17	20.7	2.95	18.75		4.25	45.25	25.80%	11.24%	18.52%
18									
19						Sum =	57.06%	57.02%	57.04%
20						Mean return (sum/5) =	11.41%	11.40%	11.41%

The following are some of the workings of the returns set out in bold in the above table for year 20.3:

$$\frac{2.00 + 16.00}{22.50} - 1 = -20\%$$

$$\frac{3.40 + 35.50}{43.75} - 1 = -11.09\%$$

$$\frac{-0.20 + -0.1109}{2} = -15.54\%$$

The deviations from the mean return for each year as well as the product of the deviations is set out in the following table in order that we may determine the variance, standard deviation for each share as well as the covariance and correlation co-efficient. This is the long way of doing it and we will also use Excel statistical functions to determine the same results.

	A	B	C	D	E	F	G	H	I
22		**Deviations from the mean return**				**Variance** [Sq. deviation]			**Covariance**
23	Year	A	B		Portfolio	A	B	Portfolio	AB
24		R_A-(avg R_A)	R_B-(avg R_B)		R_p-(avgR_p)	Deviation^2	Deviation^2	Deviation^2	dev A x dev B
25	20.3	**-31.41%**	**-22.49%**		**-26.95%**	**0.0987**	**0.0506**	**0.0726**	**0.0706**
26	20.4	8.59%	8.03%		8.31%	0.0074	0.0065	0.0069	0.0069
27	20.5	21.82%	32.21%		27.02%	0.0476	0.1037	0.0730	0.0703
28	20.6	-13.39%	-17.59%		-15.49%	0.0179	0.0309	0.0240	0.0235
29	20.7	14.39%	-0.17%		7.11%	0.0207	0.0000	0.0051	-0.0002
30									
31					Sum =	0.1923	0.1917	0.1816	0.1711
32					Variance (sum/5) =	0.0385	0.0383	0.0363	0.0342
33			Standard Deviation (sq. root of Var.) =			0.1961	0.1958	0.1906	
34									
35	***Using Excel***				*Std Deviation*	***0.1961***	***0.1958***	***0.1906***	*=STDEVP(I13:I17)*
36						*=STDEVP(G13:G17)*	*=STDEVP(H13:H17)*		
37					*Covariance*		*=COVAR(G13:G17,H13:H17)*		***0.0342***
38					*Correlation*		*=CORREL(G13:G17,H13:H17)*		***0.8914***

Page 4-35 The following are some more workings on how we get to the deviations and variance and covariance calculations for 20.3.

	E	F	G	H
55	20.3	A	B	Portfolio
56	Return	-20.00%	-11.09%	-15.54%
57	Mean return	-11.41%	-11.40%	-11.41%
58	Deviation	-31.41%	-22.49%	-26.95%

The squared deviations for Share A, Share B and the Portfolio are simply determined as follows:

$$A: (-0.3141)^2 = 0.0987 \quad B: (-0.2249)^2 = 0.0506 \quad C: (-0.2695)^2 = 0.0726$$

The covariance (product of the deviations of A and B) for 20.3 is (–0.3141)(-0.2249) = 0.0706.

Alternatively, we could also use Formulae 4.2, 4.3 and 4.4:

Correlation co-efficient:

$$\frac{0.0342}{(0.1961)(0.1958)} = 0.8907 \text{ (Rounding effect: correl} = 0.8914)$$

(b) Variance and standard deviation of portfolio:

		W	W^2	Variance	$Variance_p$	
	1	2	3	4	5	
A		0.5	0.25	0.0385	0.0096144	[3 x 4]
B		0.5	0.25	0.0383	0.0095854	[3 x 4]
		W_A	W_B	CovAB		
	2	0.5	0.5	0.0342	0.0171137	[1x2 x3 x 4]
		Variance of Portfolio			0.0363135	
		Std Deviation =		$(Var)^{(0.5)}$	0.190561	

(c) Closer to 0.9. The fact that the two shares are very similar in the direction in which returns move, results in a high correlation co-efficient, resulting in less risk reduction from investing in both than would have been the case had the correlation co-efficient been low or negative.

(d) The σ_p would decline to a point which reflects the systematic risk, that is, the risk of the market. This is certainly not zero. It would also not remain constant, as the more shares added, the greater would be the effect of less than perfect positive correlation between shares, reducing portfolio risk. If the market risk is in the region of a standard deviation of 15% (which seems reasonable), this is the most likely figure to which the portfolio risk will decline.

APPENDIX 4.1 CALCULATING THE BETA CO-EFFICIENT

Beta has been defined as the sensitivity of an individual share to changes in the index of the stock market as a whole. The market as a whole has a beta of one. Individual shares will thus have betas reflecting their relative sensitivities to the market beta of 1. An individual share, for example, which increases in price on average by 15% when the market index rises by 10% and the risk-free rate is constant, will have a beta of 1.5. Beta has a special application in the capital asset pricing model in the expression:

$$R_j = R_f + \beta_j (R_m - R_f) \qquad \textit{(Formula 4.12)}$$

where: R_f = the return expected on share *j*

Rf = the risk-free rate of return

R_m = the return expected on the market

β_j = the percentage change in an individual share's return for a 1% change in the market return

β_j, in turn, can mathematically be expressed as:

where:

$$\beta_j = \frac{Cov_{jm}}{Var_m} = \frac{\rho_{jm}\sigma_j\sigma_m}{\sigma^2_m} \qquad \textit{(Formula 4.13)}$$

$\rho_{jm}\sigma_j\sigma_m$ = the covariance of share *j*'s return with the market return (also written as cov_{jm})

σ^2_m = the variance of the market return

Modern portfolio theory suggests that covariance with the market is the only relevant risk, as unsystematic or specific company risk is virtually eliminated through diversification into a portfolio of shares. Beta reflects the covariance of an individual share with the market, standardised by dividing this covariance by the market variance. It is thus the only relevant measure of the risk of an individual share held by an investor in a portfolio.

Despite criticisms and counter evidence, the CAPM remains a widely used model. For example, services such as Bloomberg regularly publish beta co-efficients for companies listed on the JSE.

Despite the fact that betas are available for many South African shares, some analysts prefer to calculate betas based on the most recent series of price and market data. Judgement is involved in applying the techniques for calculating the beta of a share. However, the beta is in essence simply a regression co-efficient reflecting the slope of a straight line graph regressing the excess returns of an individual share against the excess return of the market as a whole. The excess return is the difference between the return and the risk-free rate, often referred to as the *risk premium*. The calculation in Table 4.11 is based on ten monthly

readings in order to demonstrate the method of calculation. In practice many more readings would be taken, and the computation would be done by computer.

Table 4.11, illustrating the calculation of beta, indicates that share J has a beta of 1.63, that is, on average, if the market index increases by 10%, the price of share J is expected to increase by 16.3%. Page 4-37

Table 4.11 Calculating beta

(1) Month-end	(2) Excess return on share J %	(3) Excess return of market %	(4) Share deviation (2) –3.6	(5) Market deviation (3) –2.7	(6) Co-variance (4) x (5)	(7) Market variance (5)²
Jan	4	3	0.4	0.3	0.12	0.09
Feb	–1	0	–4.6	–2.7	12.42	7.29
Mar	7	3	3.4	0.3	1.02	0.09
Apr	5	6	1.4	3.3	4.62	10.89
May	–4	–1	–7.6	–3.7	28.12	13.69
June	9	4	5.4	1.3	7.02	1.69
July	3	3	-0.6	0.3	–0.18	0.09
Aug	10	6	6.4	3.3	21.12	10.89
Sept	–4	–2	–7.6	–4.7	35.72	22.09
Oct	7	5	3.4	2.3	7.82	5.29
Mean	3.6	2.7			11.78	7.21

$$\beta = \frac{cov_{jm}}{\sigma_m^2} = \frac{11.78}{7.21} = 1.63$$

The procedure illustrated in Table 4.11 is as follows:

1. Record the returns on share J for each month using Formula 3.5 adapted as follows:

$$R_j = \frac{P_1 - P_0 + D}{P_0}$$

where: R_j = the excess return on share J for one month

P_1 = the share price at the end of the month

P_0 = the share price at the beginning of the month

D = any dividend which may have been paid during the month

2. Record the return on the market as a whole for each month using the overall market index as the basis, as follows:

$$R_m = \frac{OMI_1 - OMI_0}{OMI_0} + OMD_0$$

where: R_m = the return on the market as a whole for one month

OMI_1 = the overall market index at the end of the month

OMI_0 = the overall market index at the beginning of the month

OMD_0 = the overall market monthly dividend yield

3. Calculate the excess returns for both share J and the overall market by deducting the riskfree rate from the returns recorded in 1 and 2 above (columns 2 and 3).
4. Calculate the means for each set of excess returns and record the deviations from the means (columns 4 and 5).
5. Calculate the variance for the market (σ^2_m) (column 7).
6. Calculate the covariance of the return of share J with the returns of the overall market ($cov._m$) (column 6).
7. Calculate the beta in accordance with Formula 4.13.

Page 4-38 In this calculation the excess returns have been used, and dividends received on shares have been included. There will be no significant difference in the calculated beta if dividends are ignored and only price movements are used to calculate the share and market returns, because of their relative stability. Ignoring dividends and using the full return rather than the excess return is an alternative method which is used in practice.

Using Excel to determine betas

We can use Excel to determine betas. We will reorganise the data for Share J and the market as follows in Excel.

	A	B	C	D	E	F	G	H	I	J	K
3		Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct
4	Excess return Market	3.0%	0.0%	3.0%	6.0%	-1.0%	4.0%	3.0%	6.0%	-2.0%	5.0%
5	Excess return J	4.0%	-1.0%	7.0%	5.0%	-4.0%	9.0%	3.0%	10.0%	-4.0%	7.0%

We then highlight cells B4..K5, then click on **Insert** and then go to the **Charts** banner. Then select the scatter chart type. Once we have the graph, click on the data points (data points will be highlighted) and right click and then select **Add Trendline**. After that select the **Linear** option, and click on the **Display equation on chart** box and then the **Display R-squared value on chart** box. This will display the equation in the form of $y = bx + a$, where b is the slope and a is the intercept. In effect we are using regression analysis to determine a company's beta. A line that best fits the data will be inserted by Excel into the scatter graph.

It is also possible to determine the beta directly by using the SLOPE function in Excel. We can determine R-squared (R^2) by using the RSQ function and we can determine the intercept by using the INTERCEPT function.

Cell B7 = SLOPE(B5..K5,B4..K4)

Cell B8 = INTERCEPT(B5..K5,B4..K4)

Cell B9 = RSQ(B5..K5,B4..K4)

	A	B	C	D	E	F	G	H	I	J	K
3		Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct
4	Excess return Market	3.0%	0.0%	3.0%	6.0%	-1.0%	4.0%	3.0%	6.0%	-2.0%	5.0%
5	Excess return J	4.0%	-1.0%	7.0%	5.0%	-4.0%	9.0%	3.0%	10.0%	-4.0%	7.0%
6											
7	Slope	1.634									
8	Intercept	-0.01									
9	R-squared	0.828									



Page 4-39 The R-squared indicates the goodness of fit and a low R^2 indicates a high level of unsystematic or diversifiable risk. The beta may be sensitive to unique events. Assume that in our example, Share J's excess return in October is –7% rather than +7%. This will significantly impact on the beta which falls to 1.19 as compared to 1.63. The R^2 falls to 32.4%. It is important to analyse the underlying data and trends. Of course, in this example we are looking only at a few data points and this effect has been accentuated. However, the principle remains valid. Use betas with caution and apply your judgement. In the last 15 years, retailers have experienced significant increases in share prices which may reflect consolidation in the sector as well as favourable macro-economic factors. For periods of time, we will see movements in a company's share price that may be unrelated to the market due to unique or firm specific events. Let's go back to R-squared. If a company's beta is matched with a high R^2, this will indicate that the movements in the company's share price are highly dependent on movements in the market. If the beta reflects a R^2 of 1, then returns are driven only by market movements and all data points would be on the line.

A low R^2 indicates that there is very little relationship between movements in a company's share price and movements in the overall market. To put it another way, most of the changes in the company's share price reflects unsystematic risk or diversifiable risk. What we cannot diversify away is systematic risk.

Table 4.12 indicates the betas and R^2 for selected firms, sectors and indices. We would expect that sectors would depict a higher R^2.

Table 4.12 Betas and the use of R-squared to measure diversifiable risk

Sector	Beta	R^2	Selected Companies	Beta	R^2
Basic Materials	1.37	67.6%	Barloworld	0.96	25.5%
Industrials	0.86	51.6%	Anglo American	1.45	56.0%
Banks	0.94	32.6%	BHPBilliton	1.17	52.2%
Food Producers	0.32	9.2%	MTN	1.04	23.8%
ALSI TOP 40	1.03	99.0%	Tiger Brands	0.37	4.6%
Pers. Care, Drug & Grocery	0.53	30.7%	Aspen	1.02	14.0%
Healthcare	0.69	20.9%	Woolworths	0.76	18.4%

[Source: AIEMRM Equity Risk Service – June 2023]

Resources make up a significant part of the JSE and this has tended to push down other sector betas. Page 4-40

APPENDIX 4.2 PERSPECTIVES: ESTIMATING THE BETA COEFFICIENT



Obeid Mahomed offers us his perspectives on estimating beta coefficients in South Africa. This is particularly relevant for practitioners. Obeid is a lecturer in Mathematical Finance, Risk Management and Financial Technology at the African Institute of Financial Markets and Risk Management (AIFMRM) at the University of Cape Town. His previous work experience was in the financial services sector, with extensive exposure to banking, asset management and broking.

Introduction

The genesis of the beta concept is the Capital Asset Pricing Model[10] (CAPM), which is an equilibrium model that purports relations between expected returns on risky assets with the beta parameter being the fundamental relational quantity. The estimation of the beta parameter has traditionally been achieved by running an Index or Market Model regression. Running this regression however can lead to a variety of practical considerations which in turn could result in several different estimates for beta. Some of these considerations could be purely measurement related such as: How does one measure returns? What market proxy should be used? How long should the return intervals be? How many data points are needed?

On the other hand, a further set of considerations involve the assumptions and the inferences, such as: Is thin trading a problem? Is the market segmented? Are betas likely to be stable?

This appendix reviews some of the procedures that need to be considered to ensure the resulting betas are accurate. For users of published betas, this appendix gives readers an understanding of the care needed in the estimation of beta coefficients.

The Market Model

In its simplest form, and in the context of equity markets, the beta parameter describes the extent to which a share's returns follow the returns in an equity index that is representative of the overall market. Clearly some stocks tend to move more than others when the market moves; hence their sensitivity to movements in the overall market index is an important measure. This sensitivity is quantified by beta and widely referred to as the beta coefficient. The following Market Model has traditionally been used to estimate the beta coefficient:

$$R_{it} - R_{ft} = \alpha_i + \beta_i (R_{mt} - R_{ft}) + e_{it} \quad (1)$$

where R_{it} is the periodic return on share i over the period $[t-1, t]$;

R_{ft} is the periodic risk-free return over the period $[t-1, t]$;

R_{mt} is the return on the equity market index over the period $[t-1, t]$;

α_i and β_i are the intercept and slope (beta) coefficients to be estimated for asset i;

e_t is a measure of idiosyncratic or asset-specific risk for share i over the period $[t-1, t]$;

the length of the period $[t-1, t]$ is typically a day, week, month, or quarter; and

$t \in \{1,2,\ldots, T-1, T\}$ denotes a counter, with T being the size of the total sample.

The Market Model, given by equation (1), is a simple linear regression model and may therefore be resolved/estimated using ordinary least squares (OLS). In this instance the OLS estimate of beta is simply:

$$\beta_i = \frac{\sum_{t=1}^{T}(x_t - \bar{x})(y_{it} - \bar{y}_i)}{\sum_{t=1}^{T}(x_t - \bar{x})^2} \quad (2)$$

Page 4-41 where $x_t = R_{mt} - R_{ft}$ is the excess periodic market return over the period $[t-1, t]$;

$\bar{x} = \frac{1}{T}\sum_{t=1}^{T} x_t$ is the average excess period market return over the entire sample period $[0,T]$;

$y_{it} = R_{it} - R_{ft}$ is the excess periodic return on share i over the period $[t-1, t]$; and

$\bar{y}_i = \frac{1}{T}\sum_{t=1}^{T} y_{it}$ is the average excess period return on share i over the entire sample period $[0.T]$.

It should be noted that the market model is not based on any assumptions about investment behaviour but simply posits a linear relationship between stock and market excess returns. Figure 4.16 and Figure 4.17 show the scatter diagrams for two stocks Anglo American plc (AGL) and Shoprite Holdings Ltd (SHP) regressed on the All Share Index (ALSI) respectively. It is evident that the estimated (OLS) beta for AGL is 1.47 and for SHP is 0.24, indicating their differing sensitivities to market movements. It should be noted that whilst this appendix focuses on the estimation of betas for individual stocks, the concept could be applied to any asset, including investment portfolios.



Figure 4.16 Scatter diagram for Anglo American plc

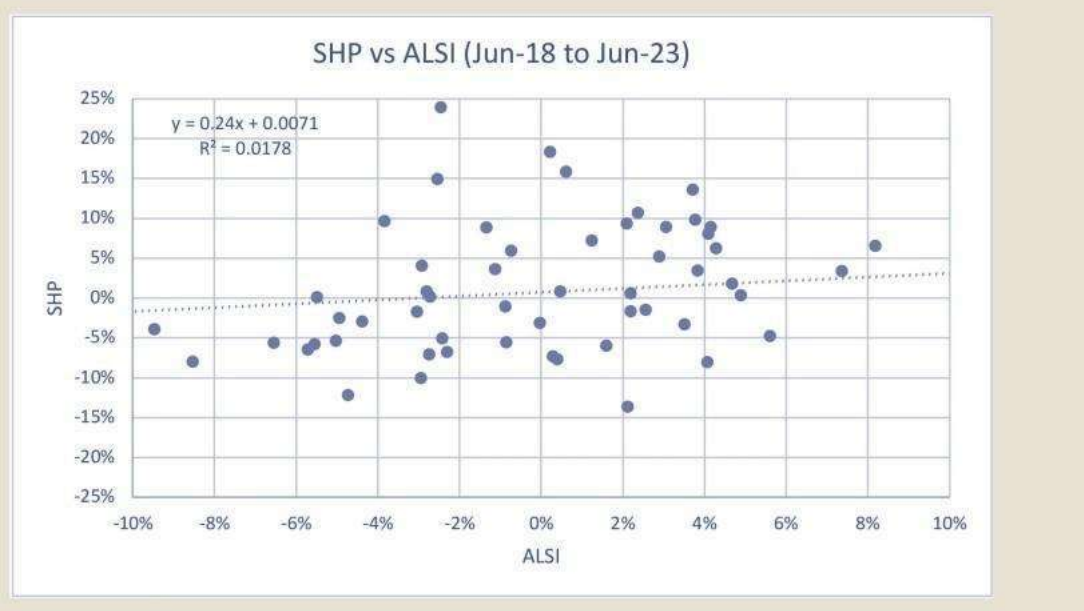


Figure 4.17 Scatter diagram for Shoprite Holdings Ltd

Depending on the application, one may choose to use either *discrete or continuously compounded* returns in the Market Model presented above, as long as the consistency between the share, risk-free returns and market index proxy is maintained. It is generally accepted that returns are continuously generated through calendar time, however because trading occurs at discrete intervals, observers view returns as if they are generated at discrete intervals. For both the discrete and continuously compounded returns it is important to note that the returns should be adjusted for capitalization changes and dividends.

The Market Index

In theory, market capitalization weighted indices are preferred to equally weighted indices because they are superior proxies to the true market portfolio. Hence, in South Africa, it is generally accepted that the All Share Index (ALSI) should be used. It should be noted that some practitioners argue that there is a segmentation amongst the Resources, Financial and Industrial sectors on the JSE and consequently prefer to use these sub-sector indices as an overall market proxy for stocks belonging to these sectors. It should be noted however that the standard theory calls for an index that is as comprehensive as possible in relation to covering the market.

Length of the estimation period

Estimates based on many years of historical data may be of little relevance because the nature of the business risks undertaken by companies may have changed significantly over a long period such as 10 years. It is generally accepted that a 5-year estimation period is viable. The choice of a 5-year estimation period is based on the findings that betas tend to be reasonably stable over 5 yearly periods. The selection of a 5-year period represents a satisfactory tradeoff between a large enough sample size to enable reasonably efficient estimation and a short enough period over which the underlying beta could be assumed to be stable.

The return interval

Academic research has been directed at establishing the impact that different interval lengths have on estimates of beta. As a consequence, subsequent researchers generally use monthly intervals (over a 5 year period) to compute the returns needed for the estimation process, resulting in 60 data points of monthly returns.

Correcting for the regression bias using a Bayesian adjustment

A number of studies have documented that individual stock betas have a regression tendency towards the grand mean of betas of all stocks on the exchange. This regression bias can be described as follows: an estimated beta coefficient which is far higher than the average beta is more likely to be an overestimate of the true beta than an underestimate. Similarly, a very low estimated beta is more likely to be an underestimate. Thus, the estimates of beta obtained from the regression analysis may be suboptimal for forecasting purposes. To correct for the

regression bias, a "Bayesian" adjustment can be implemented. Hence betas may be corrected as follows:

$$\hat{\beta} = w\beta_{OLS} + (1-w)\bar{\beta}_{OLS} \quad (3)$$

$$w = \frac{\sigma_{\beta}^2}{(\sigma_{\beta}^2 + \sigma_{OLS}^2)} \quad (4)$$

where β_{OLS} is the OLS estimate of beta

$\bar{\beta}_{OLS}$ is the average OLS beta of all stocks in the market

σ_{OLS} is the standard error of the OLS estimate of beta

σ_{β} is the cross-sectional standard deviation of all the estimates of beta in the market Page 4-43

It can be readily seen from the above formulation (4) that the weight, *w*, assigned to the OLS beta will be large if the cross-sectional standard deviation (σ_b), is large. In other words, if the spread of the betas across the stocks is so large as to make all values of beta equally likely, then the OLS beta estimator is optimal. Conversely, if the standard error of the OLS beta estimate, σ_{OLS}, is large relative to σ_{β}, then *w* will be small and the Bayesian estimate of beta will be "shrunk" towards the overall average beta of the stocks on the market. Therefore, an estimate of beta which falls outside the usual spread of beta (and which has a large standard error) is likely to be an overestimate. Hence the above expression corrects for the commonly observed phenomenon that very high beta estimates that are unreliable (large standard errors) tend to be overestimates and very low betas that are unreliable tend to be underestimates.

Adjusting for thin-trading

The bias in beta estimates caused by thin-trading on the JSE has been well documented. If a stock is thinly traded, then it is likely that the month-end price may not arise from a trade on that day but may instead be recorded as the price last traded during the month. Consequently, the recorded price on the market index at month-end may not be matched to a trade for the stock on the day – hence a mismatch occurs. This mismatching phenomenon clearly has an impact on the covariance estimate between the stock and the market proxy, leading to a bias in this covariance estimate and consequently a bias in the estimate of beta.

Several researchers have derived techniques for obtaining unbiased estimates for beta in infrequently traded environments. Two distinctly different approaches have emerged; the "trade-to-trade" estimator and the Cohen estimators. It was found that the trade-to-trade was superior for application on the JSE. In the trade-to-trade method, the returns on the stock and on the index are measured between the times of the last trades in successive months. Thus, a statistical correction needs to be made for potential durations in return no longer being equal.

Furthermore, to improve the efficiency a correction is required for the heteroscedasticity in the residual component – this leads to the final trade-to-trade estimator proposed by Dimson and Marsh:

$$\frac{R_{iu} - R_{fu}}{\sqrt{D_{iu}}} = \alpha_i \sqrt{D_{iu}} + \beta_i \frac{R_{mu} - R_{fu}}{\sqrt{D_{iu}}} + e_{iu} \qquad (5)$$

where the counter $u \in \{1,2, \ldots, U\}$ still counts the number of data points within a sample, in this case U, but now the length of each period depends on the last day a trade occurred in relation to the chosen interval; and D_{iu} is the length of the period $[u - 1, u]$, expressed as fractions of a year, or in other words the length of time between successive trades.

The rest of the quantities are as defined in the standard Market Model, given by equation (1).

Other issues relating to the use of betas

The stability of betas

Detection of beta stability is clouded by the fact that only the estimates of beta are observable. Changing estimates do not necessarily imply stable underlying betas. Research indicates that the results of tests on the stability of betas are difficult to interpret but betas for JSE-listed shares were found to be as stable as betas of stocks listed in the UK equity market.

Robust estimation of beta

Research indicates that the robustness required for estimating betas involved down-weighting not only the outlying residual values, but also the outlying market returns. Page 4-44

Dynamic estimation of beta

Filtering and smoothing techniques can serve as useful aids in the interpretation of time series of return observations. Techniques are available that categorize the movements in the levels of stock returns as either permanent shifts in the riskiness of the firm, or as one-off events attributable to specific circumstances.

Using a "beta book"

There are many services that supply betas and associated estimates arising from the Market Model. We have included below an extract from the Equity Risk Service (June 2023) that is published by AIFMRM on a quarterly basis. This is one of the most widely used beta services in South Africa (according to a PWC survey). This service uses both the trade-to-trade correction procedure given in equation (5) as well as the Bayesian correction procedure given by equation (3).

Table 4.13 Extract from Equity Risk Service Beta book

FTSE/JSE All Share Index (J203) as Market Proxy

Code	*Name*	*Start Date*	*End Date*	*Days Traded*	*Data Points*	*Alpha*	*Beta*	*SE Alpha*	*SE Beta*	*R^2*	*Total Risk*	*Uni Ri*
ABG	Absa Group Limited	29-jun-18	30-jun-23	100%	60	0.00078	1.20739	1.13%	21·02%	31.80%	36.80%	30.2
AGL	Anglo American plc	29-Jun-18	30-Jun-23	100%	60	0.00724	1.44959	0.80%	15.57%	55.95%	33.14%	21.9
AMS	Anglo American Plat Ltd	29-Jun-18	30-Jun-23	100%	60	0.01786	1.57432	1.50%	25.98%	29.96%	49.83%	41.7
ANG	Anglogold Ashanti Ltd	29-Jun-18	30-Jun-23	100%	60	0.02691	1.06791	1.96%	31.55%	10.96%	55.14%	52.0
ANH	Anheuser-Busch InBev	29-Jun-18	30-Jun-23	100%	60	0.00667	0.78270	0.93%	17.74%	23.28%	27.74%	24.3
BHG	BHP Group Limited	29-Jun-18	30-Jun-23	100%	60	0.00692	1.16742	0.72%	14.00%	52.19%	27.63%	19.1
BID	BID Corporation Ltd	29-Jun-18	30-Jun-23	100%	60	0.00418	1.14117	0.79%	15.42%	45.79%	28.84%	21.2
BTI	British American Tob plc	29-Jun-18	30-Jun-23	100%	60	-0.00372	0.46133	0.96%	18.24%	9.03%	26.25%	25.0
CFR	Compagnie Fin Richemont	29-Jun-18	30-Jun-23	100%	60	0.1507	1.26071	0.93%	1.7.82%	42.38%	33.11%	25.1
CPI	Capitec Bank Hldgs Ltd	29-Jun-18	30-Jun-23	100%	60	0.00947	0.83247	1.19%	21.93%	17.28%	34.25%	31.1
FSR	Firstrand Ltd	29-Jun-18	30-Jun-23	100%	60	0.00106	0.82731	0.92%	17.56%	25.78%	27.86%	24.0
GFI	Gold Fields Ltd	29-Jun-18	30-Jun-23	100%	60	0.03619	1.15804	2.14%	33.24%	10.75%	60.38%	57.0
GLN	Glencore plc	29-Jun-18	30-Jun-23	100%	60	0.00651	1.45741	0.91%	17.40%	50.24%	35.16%	24.8
IMP	Impala Platinum Hlgs Ltd	29-Jun-18	30-Jun-23	100%	60	0.03769	1.45362	2.02%	32.11%	16.82%	60·61%	55.2

FTSE/JSE All Share Index (J203) as Market Proxy												
KIO	Kumba Iron Ore Ltd	29-Jun-18	30-Jun-23	100%	60	0-00898	1.32465	1.52%	26·50%	23.29%	46-93%	41.1
MNP	Mondi plc	29-Jun-18	30-Jun-23	100%	60	-0.00743	0.71363	0.77%	15.00%	26.39%	23.53%	20.1
IMTN	MTN Group Ltd	29-Jun-18	30-Jun-23	100%	60	0.00407	1.03728	1.20%	22.11%	23.82%	36.34%	31.7
NPN	Naspers Ltd -N-	29-Jun-18	30-Jun-23	100%	60	0.00733	0.64052	1.34%	24.06%	8.92%	36.67%	35.0
PRX	Prosus N.V.	30-Sep-19	30-Jun-23	100%	45	0.00491	0.58053	1.67%	26.93%	7.53%	38.89%	37.4
S32	South 32 Limited	29-Jun-18	30-Jun-23	100%	60	0.00272	1.10749	1.00%	18.99%	33.65%	32.64%	26.5
SBK	Standard Bank Group Ltd	29-Jun-18	30-Jun-23	100%	60	-0.00323	0.88541	0.95%	18.13%	26.92%	29.18%	24.9
SHP	Shoprite Holdings Ltd	29-Jun-18	30-Jun-23	100%	60	-0.00043	0.39842	1.15%	21.32%	4.89%	3.82%	30.0
SLM	Sanlam Limited	29-Jun-18	30-Jun-23	100%	60	-0.00562	1.04844	0.81%	15.65%	41.19%	27.93%	21.4
SOL	Sasol Limited	29-Jun-18	30-Jun-23	100%	60	0.01003	2.28794	2.461%	35.90%	20.12%	87.20%	77.9
VOD	Vodacom Group Ltd	29-Jun-18	30-Jun-23	100%	60	-0.00408	0.25606	0.67%	13.17%	5.86%	18.08%	17.5

Table 4.13 represents the output from a typical beta service. The columns give statistics emerging from Market Model regressions, the following interpretations are relevant:

- *Alpha*: This is the average return (per month) on the share when the market on average does not move.
- *SE Alpha*/*Beta*: The standard error of alpha/beta is a statistical measure of the reliability of the estimate of alpha/beta. The lower this figure, the more reliable the estimate of alpha/beta. Statisticians set up confidence intervals for the estimate of alpha/beta by

adding and subtracting *2 x SE Alpha/Beta* from the alpha/beta estimate. There is approximately a 95% chance that the true alpha/beta lies in this interval.

- *Total Risk*: This is the annualized standard deviation of returns measuring the share's total risk.
- Unique Risk (or Specific Risk): This reflects the fluctuations in the security's returns that are linked to events unique to the company (e.g., bad management, worker strikes etc.).
- R^2 : This can be interpreted as the proportion of the share's total risk accounted for by its market risk. Note that a high beta will not necessarily produce a high R^2. In statistical terms, R^2 is the coefficient of determination of the regression.
- *Start Date & End Date*: These specify the first and last month in the period of analysis for a specific company – this varies depending on when the respective company listed on the exchange.
- *% Days Traded*: The number of days over the period of analysis during which the security was traded. This provides an indication of the extent to which the security is thinly traded. Page 4-45 In some instances, for extremely thinly traded securities, more than 5 years of data may be needed in order to obtain a data set that is large enough to warrant statistical estimation.

Summary[11]

The primary aim of this appendix has been to provide guidance to the practitioner wanting to estimate betas. Research in South Africa has indicated significant bias in beta estimates caused by thin trading as well as the tendency for mean reversion. Hence, of all the refinements briefly discussed, the most important of these are the thin-trading correction (especially when it is known that stocks suffer from thin-trading) and the Bayesian correction.

APPENDIX 4.3 PERSPECTIVES: BEHAVIOURAL FINANCE[11]



Darron West (MCom MPhil LLB CA(SA) CFA) is a practising advocate, member of the Cape Bar and a part-time senior lecturer in the Department of Finance and Tax at the University of Cape Town. He continues to consult to the investment management industry, and he is a non-executive director of a South African short-term insurance company. Since his career started in financial services in 1998, he has watched behavioural finance at work in financial markets. His experience spans derivatives, structured investment products, fund management, venture capital, non-profit organisations and insurance.

Classical finance theory teaches us the proverbial efficient markets hypothesis (EMH). In a nutshell, the EMH postulates that prices on traded assets reflect all known information; hence, such prices are unbiased. Consequently, it is impossible to consistently outperform the market on a risk-adjusted basis except through sheer luck. Investors seek to maximise their utility, and as such they have rational expectations (on average, but for every lunatic there is an offsetting sane and prudent participant).

However, even a casual analysis of the price series of an equity market compared with its rational valuation (the present value of discounted cash flows) illustrates that the unexplained component of the efficient market models is substantial. Why is it that at the late stages of a bull market, prices appear to be driven up withouxt any consideration of value? Why is it that at the end of market crashes, prices appear to be in free fall as investors exit their holdings irrespective of value? If EMH held, surely participants should always take advantage of fundamentally mispriced securities?

The fact is, they don't. The evidence suggests market overreaction is beyond the realms of rationality. Price is not equal to the present value of discounted cash flows, risky assets are not necessarily rated lower than riskless assets and markets are not reasonably efficient most of the time. The reality is that market participants are not automatons. Programme trading aside, market participants are human beings with all the foibles and trappings of that condition. Truly understanding the financial markets must involve an understanding of investor psychology and the associated (and often systematic) non-rationality that follows. This does not imply that EMH should be dismissed; rather, it connotes an attempt to have a more complete understanding of market behaviour.

It would be trite to suggest that behavioural finance is merely the incorporation of psychology into finance, but it certainly involves specific consideration of human and social cognitive and emotional biases and dispositions in better understanding how and why economic decisions are made, and how these decisions affect market prices, returns and the allocation of resources. A simple test will reveal a differential treatment of risk. Given a choice between (1) a 100% chance of winning R1m, and (2) an 89% chance of winning R1m, a 1% chance of winning nothing and a 10% chance of winning R5m, most people would choose (1), even though the rational probability weighted payoff of (2) is higher. However, when faced with a choice between (3) an 11% chance of winning R1m and (4) a 10% chance

of winning R5m, most people would choose (4). The rational utility preference is simply not consistent, and the cognitive impact of certainty or near-certainty is far higher than perceived remote or unlikely events. The experiment above is an exposition of prospect theory, coined seminally by Kahneman and Tversky in 1979 (for which Kahneman was awarded the Nobel Prize for Economics in 2002). Prospect theory shows how human beings have a concave utility curve for gains, and a convex utility curve for losses. More bluntly, we feel the pain of losses (even small ones) from a reference point far more acutely than we feel the joy of gains. Many of the observed behaviours and biases inherent in human participation in the financial markets flow from this simple basis.

Once sensitised to them, it is hard not to observe these behavioural biases in ourselves and in others. Most revolve around the perception and understanding of loss. Because it is painful to make a mistake, we attempt to avoid regret by altering the basis of evaluation so as to change an outcome from failure to 'deferred success'. Separating actual losses from paper losses, holding on to losing stocks and selling winners early are all symptoms. Similarly, as the fabled (and, indeed, fictional) ostrich, we figuratively bury our heads in the sand when confronted with new information that would challenge our beliefs or assumptions (and so force a reversal of the action we have taken, or induce regret). This 'cognitive dissonance' manifests broadly, from avoiding advertisements for newly purchased products (in case they are cheaper elsewhere), to avoidance of information about a new investment (in case it proves to be misguided).

Research has shown how we are influenced by reference points (anchoring) in determining outcomes, we fail to view portfolios on an aggregate basis (creating mental compartments), we create the illusion of patterns where none exist and attribute losses to bad luck (the problem of overconfidence), and we fit hypotheses to data instead of considering probabilities more rationally (the representativeness heuristic). In fields requiring some prospective estimation to justify decisions, overconfidence has been shown to be a perennial failing. Forecasts have been shown to be both too high and hardly useful at predicting turning points. Furthermore, studies examining gender differentials have shown that women are less overconfident, trade less (thus incurring lower costs) and produce better returns than men (which begs the question why the fund management industry remains so male dominated!).

Is behavioural finance merely a tool for poking the EMH full of holes? Hardly, challenging the EMH is a starting point, as is using the knowledge of human behavioural biases to guard against prejudicial or damaging financial conduct. The real power of behavioural finance has still to be harnessed, at the very least to fill the gaps between EMH models and actual market outcomes, and perhaps as a grand unifying theory of finance.

APPENDIX 4.4. INVESTMENT DECISION-MAKING UNDER CONDITIONS OF UNCERTAINTY



Ameer Amod is Head of Absolute Return and Fixed Interest at Old Mutual MultiManagers. Ameer began his career in Financial Services in 2002 and has gained extensive industry experience, working for Metropolitan Asset Managers as a Portfolio Manager, Futuregrowth Asset Management as the Head of Quantitative Research and Peregrine Quant Asset Management as a Quantitative Analyst. He has an MCom (Financial Management) and an MSc (Engineering). Ameer joined Old Mutual in January 2012. He is a published writer and regularly presents at investment conferences and to clients.

Financial markets essentially aggregate human behaviour. Cognitive biases cause investors to produce predictable mispricing in financial markets. Research has shown that individuals are much more sensitive to losses in wealth than to gains in wealth – in fact, twice as much. The adage 'crisis creates opportunity' fails to be exploited when valuation opportunities emerge in financial markets. Former Fed Chairman, Alan Greenspan, when referring to Black Monday puts it succinctly '*your judgement is less than perfect when you are scared*'. We tend to become less sensitive to valuation when things are getting bad – and when things get bad, it is human nature to exaggerate the risks. We highlight a few useful concepts that we as students of finance need to give due consideration when making asset allocation decisions.

TIME HORIZON

The asset allocation decision is independent of time horizon (Merton, 1969). This has always been controversial and counterintuitive – investment advisers explicitly recommend changing the asset mix toward more conservative investments as retirement becomes closer. At retirement, depending on an individual's savings to income ratio[12], assets yielding 1%–2% above inflation may not be sufficient to fund increases in future cost of living. Benartzi & Thaler (1999) have found that retirement fund members are likely to invest *more of their retirement savings in equities* rather than *cash* when shown longer-term rates of return. Long term should be consistent with the length of time a member would be part of a retirement fund or some long-term savings plan. This gives the fund member time to accumulate the higher returns generally expected from growth asset classes like equity.

THE EQUITY RISK PREMIUM ('ERP')

There is an enormous discrepancy between the returns of stocks and bonds – denoted the 'equity risk premium'. Historically, the average return on equity has been much larger than that of bonds. Firer & Staunton (2002) have shown this ERP to be 5.4% for South Africa over

the period spanning 1900–2001. The period 1900 to 2022 indicated a real ERP of 5.1% for South Africa (Credit Suisse Global Investment Returns Yearbook, 2023). To quantify and interpret the magnitude of this ERP, it is instructive to examine the US market, where we have a rich history of data. Siegel (2023) examined the ERP for the US over the period 1802–2021 as follows:

PERIODS	STOCK MARKET RETURNS	STOCKMARKET REIURNS	LONG TERM GOVERNMENT BONDS	LONG TERM GOVERNMENT BONDS	SHORT TERM GOVERNMENT BILLS	SHOI GOVE B
	Nominal	Real	Nominal	Real	Nominal	
1802 – 2021	8.4%	6.9%	5.0%	3.6%	4.0%	
1802 – 2012	8.1%	6.6%	5.1%	3.6%	4.2%	
1802 – 1870	6.9%	6.7%	4.9%	4.8%	5.2%	
1871 – 1925	7.3%	6.6%	4.3%	3.7%	3.8%	
1926 – 2012	9.6%	6.4%	5.7%	2.6%	3.6%	

Page 4-48 The 2023 Credit Suisse Investment Returns Yearbook reported a real ERP of 4.6% for the USA over the period 1900 to 2022. This reflects the strong performance of US equity markets in recent years. The US has transitioned from an agricultural to an industrialist economy, then to the postindustrial service and technology economy it is today. Interestingly, over these transitions the real returns from equities has been remarkably stable.

Investors today have more reason to ask what return they can expect from equities and bonds; and what the future risk–reward trade-off is likely to be. The magnitude of the ERP is one of the most important issues in corporate finance. It drives future returns and is a key determinant in the cost of capital. Blanchard (1993) and Campbell & Shiller (1998) believe that it is unreasonable to assume that the ERP will remain as large as historically observed. Siegel (1992) explains that the higher equity risk premium is a result of the fall in bond returns rather than a rise in the return on equity. The move away from the gold standard to paper money has had a far greater impact on bond returns than on equities. Buyers of long-term bonds in the 1940s, 1950s and early 1960s did not anticipate the inflationary consequences of this shift in monetary regime. It is therefore not unreasonable to believe that low real rates on bonds may have fuelled higher equity returns supported by the low cost of leverage. The ERP is not nearly as large as historically observed versus the post-1926 period. With many central banks targeting inflation, holders of fixed income assets are likely to see

enhanced real returns in the future – however, equities still dominate as an asset class to long-term wealth creation.

VALUATION

Value is subjective and inherently uncertain – decisions made in most real-world applications are in the presence of uncertainty or incomplete information. The price paid for an asset today determines its future return. Euphoria in financial markets has led investors to pay high prices for assets – if the future is not as rosy as contemplated by these investors, future returns will be disappointing. Invest when expectations incorporated into asset prices are prone to extreme pessimism; when things turn out to be 'relatively ok' we have the benefit of a surprise to the upside. Howard Marks[13] suggests that the action of an investor should be governed by the relationship of an asset's price to its intrinsic value. When assets are cheap investors should be more aggressive in buying and conversely when assets are expensive we should exercise restraint. Key questions to be deliberated are (i) is there a sufficient margin of safety in the current price, and (ii) are fundamentals likely to improve or get incrementally worse?

We have discussed a few critical considerations when thinking about the asset allocation decision. The issue of time horizon, the consideration of whether the equity risk premium is likely to shrink in future and, most importantly, the price we are willing to pay for an asset today. The growing area of behavioural finance has also begun to improve our understanding of the impact of investor psychology on asset prices. We urge readers to deeply consider these issues when making investment decisions.

APPENDIX 4.5 PROSPECT THEORY Page 4-49



Dr Gizelle Willows is a Professor in the College of Accounting at the University of Cape Town and a Chartered Accountant (SA). Gizelle's PhD was on behavioural investing and retirement planning and she has published widely in peer-reviewed journals. She has been a visiting scholar at the Haas School of Business, University of California, Berkley. She published a column in Accountancy SA for many years. Prior to joining UCT, Gizelle was an audit manager at PWC.

Daniel Kahneman and Amos Tversky developed a theory known as prospect theory, which shows that as individuals, we make decisions based on the value of potential losses and gains, rather than the outcome. Figure 4.18 helps to explain it more clearly:

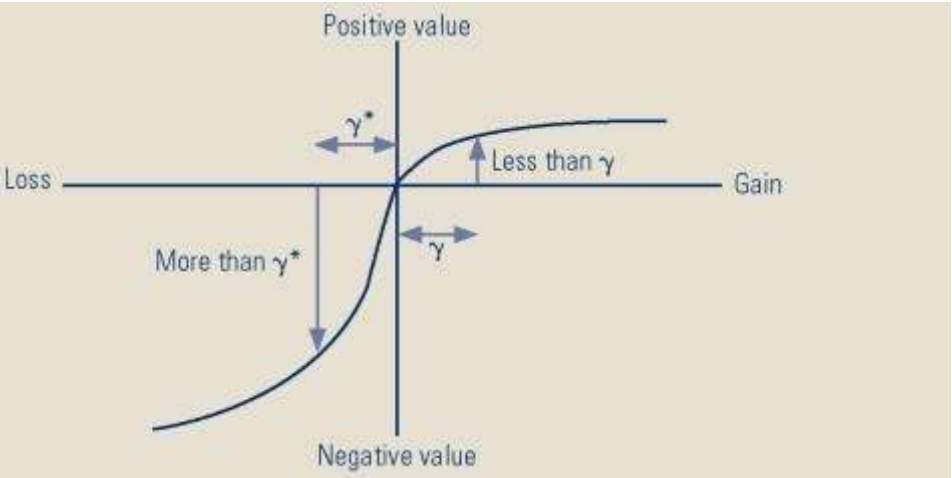


Figure 4.18 Prospect Theory

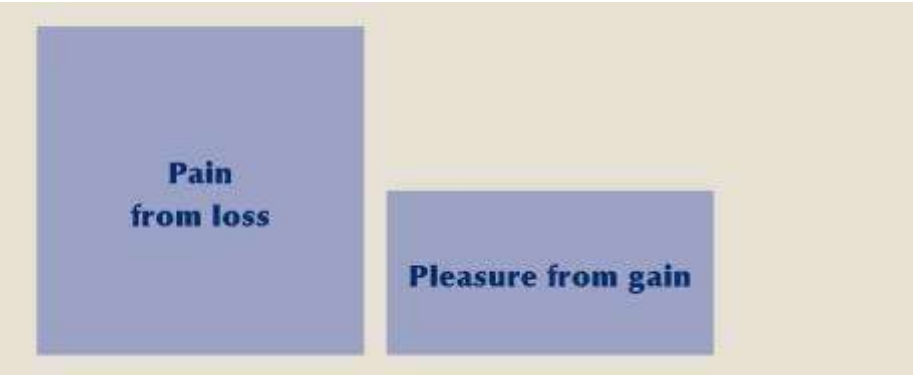
The horizontal axis measures the value of our gain or loss, whilst the vertical axis measures how we value that gain or loss. Looking at the horizontal line (shown as 'γ') on the right-hand side of the graph, it shows the objective value of a certain gain. The vertical arrow pointing upwards reflects the subjective value we attach to this gain. The observation to take note of is that the subjective value we place is less than the actual objective gain.

Looking at the other horizontal arrow (measured at the same value, i.e. 'γ*') on the left-hand side of the graph, it shows the same value, but this time as a loss. The vertical arrow pointing downwards is longer than the horizontal arrow, showing that the subjective value we attach to this loss is much larger than the actual loss itself.

If a gain or a loss was R1 000 it shows that we place a larger value on a loss of R1 000, than on a gain of R1 000. Prospect theory shows us that we are loss averse i.e. risk-averse over gains and risk-seeking over losses and centred to a set reference point or status quo. This reference point is subjective and can be different for different people in different scenarios.

In 1974, Amos Tversky and Daniel Kahneman performed an experiment where they asked participants: 'what is the percentage of African nations in the UN?' while spinning a wheel of fortune with numbers from 1 to 100 on it. Where the wheel stopped was completely irrelevant, however, it was found that it influenced the answers given by the participants. This study showed how a completely irrelevant number can affect our decision-making. This behaviour is also referred to as anchoring, a suitably named bias. Anchors of ships are put into the ocean to ensure they don't stray too far away from a certain point. We do the same with our reference points.

A good example of a product designed to take advantage of the behaviours attached to prospect theory are insurance policies. If we are loss-averse and make decisions to avoid potential losses, then insurance products are perfect for us. They are designed to highlight the potential loss. This then creates the anchor pushing us to choose the option that avoids such losses[14].



To summarise, prospective losses bother individuals much more than prospective gains. Therefore, the choices we make are often based on the subjective version of a situation, rather than the objective reality. We should be mindful of this and always differentiate between price and value. Make decisions based on the intrinsic value of the transaction.

QUESTIONS

Question 4.1

The following summary statistics are available with regard to shares A, B and C:

- Expected returns:R_a=10%;R_b=13%;R_c=20%
- Standard deviations:σ_a=8%;σ_b=15%;σ_c=25%
- Correlation co-efficient:r_{ab}=0.6;r_{ac}=0.8;r_{bc}=–0.4

Required:

(a) Determine which of shares A, B, and C would be the most attractive to a risk-averse investor who uses the mean–variance criterion to minimise the risk per unit of return. A ranking of shares A, B, and C is required.

(b) An investor intends to hold a portfolio of investments comprising two of the three shares A, B and C (equal amounts would be invested in each of the two shares), therefore the following options are available:

- Option 1: 50% in each of shares A and B
- Option 2: 50% in each of shares A and C
- Option 3: 50% in each of shares B and C

Determine which of the three options would be the most attractive to a risk-averse investor. A ranking of the options is required.

Question 4.2

Two shares, P and Q, offer the following four historical percentage annual returns:

Shares	1	2	3	4
P	10%	12%	5%	13%
Q	20%	6%	10%	12%

An investor decides to hold a portfolio of 40% invested in share P and 60% invested in share Q.

It is evident that the calculations will suffer from small sample bias. Nevertheless, you are required to demonstrate the procedures used for the calculations and your ability to interpret the results.

Required:

(a) Calculate the correlation co-efficient of shares P and Q.

(b) Calculate the expected return on the portfolio.

(c) Calculate the variance of the portfolio.

(d) The investor believes that the risk of a portfolio containing any combination of P and Q will always be less than the risk of holding either P or Q alone. Discuss whether you agree or disagree with this view, giving reasons. Page 4-51

Question 4.3

An investment of R500 000 held by your company at the riskless rate of return of 8% has just matured. It has been decided to invest the proceeds in ordinary shares listed on the JSE. Two shares, A and B, have been identified with a correlation co-efficient between A and B (p) of zero, and you can invest the full R500 000 in either of them, or in a portfolio with some of each.

The following facts are available regarding the two shares:

	Share A	Share B
Expected return	10%	14%
Standard deviation of returns	10%	15%

Required:

(a) Draft a table which shows 'expected portfolio % return' and '% standard deviation of portfolio return', given that the five following portfolios are considered: Investment in share A (the balance in share B) – 100%; 75%; 50%; 25%; 0%.

(b) Sketch a graph that plots the five portfolios in terms of expected risk and return. Label points that indicate:

(i) The feasible set of portfolios.

(ii) The efficient set of portfolios.

(iii) The indifference curves of an extremely risk-averse investor, and

(iv) The indifference curves of a moderately risk-averse investor.

Question 4.4

The following limited information on annual returns is available for two shares listed on the Johannesburg Securities Exchange.

Year	Hifli	Lowfli
20.1	18	16
20.2	22	12
20.3	36	10
20.4	12	18

Despite the limited number of readings, a normal distribution of returns may be assumed. In addition, past performance is considered to reflect expected future performance.

Required:

(a) Calculate the average return and standard deviation of each share.

(b) Calculate the covariance of returns for a portfolio comprising shares in both Hifli and Lowfli.

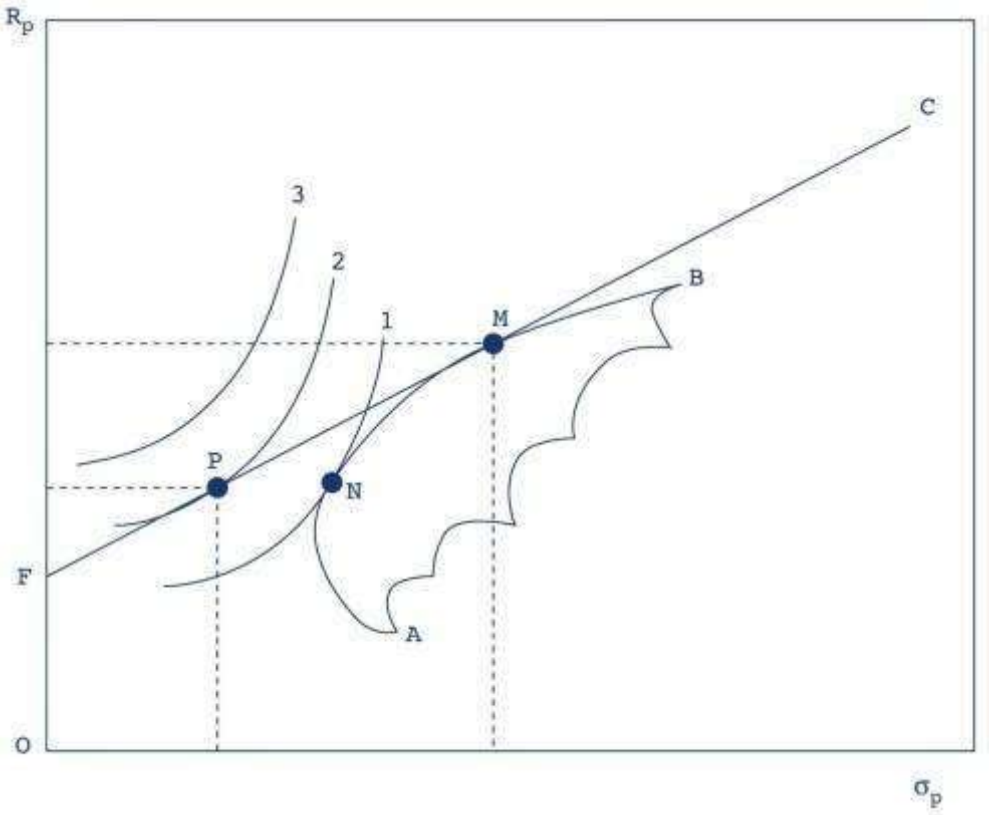
(c) Calculate the correlation co-efficient of returns in a portfolio of shares of Hifli and Lowfli.

(d) Assume you have R10 000 available to invest. If you wanted to buy either Hifli or Lowfli, which one would you buy? Explain your conclusion.

(e) If you choose to invest R3 000 in Hifli and R7 000 in Lowfli, calculate your expected return from the portfolio and a measure of risk.

(f) Explain briefly the rationale for investors choosing to hold portfolios of shares rather than individual shares.

Question 4.5



Required:

Referring to the diagram, you are required to discuss the significance of each of the following:

(a) curve ANMB

(b) curves 1, 2, 3

(c) line FC

(d) point A

(e) point B

(f) point F

(g) point M

(h) point N

(i) point P

Question 4.6

Illustrate the following with a diagram or graph, ensuring that relevant lines, points, and axes are clearly labelled (no values need be placed on axes, and graphs need only be sketched):

(i) the probability distribution of returns on two individual shares with different expected returns;

(ii) opportunity set of share portfolios and the capital-market line. Indicate with an L the positioning of a leveraged portfolio;

(iii)the effect on risk of holding an increasing number of individual shares in a portfolio;

(iv)the security-market line – indicate with an R the positioning of a share with a higher-thanaverage market risk; and

(v) approximate regression lines for four shares with beta values of 3, 1, 0.4, and –0.4 respectively.

Question 4.7

Ms Lightfoot wishes to invest 50% of her savings in shares of Company A and 50% in shares of Company B. The expected returns and standard deviations for A and B are as follows:

Company	Expected Return	Standard
		Deviation of returns
A	10%	15%
B	15%	20%

Required:

(a) What is the expected return of the portfolio?

(b) What is the variance and standard deviation of returns of the portfolio if the correlation between the returns of A and B is: Page 4-53

a. +1.0

b. +0.0

c. -0.8

(c) Assume that she wishes to invest 40% in Company A and 60% in Company B. How would this change the expected return of the portfolio? Recalculate the variance and standard deviation of the portfolio's returns, if the correlation of returns is zero.

Question 4.8

As an investor, Ms Malinga is considering investing in shares rather than bonds. This is due to the recovery in equity markets. She is considering investing in the shares of three companies, Company X (a utility), Y (a retailer) and Z (a mining company). The expected return on the market is 12% and the risk-free rate is 7%. She wishes to use the CAPM to estimate the required rate of return for the three shares. The beta for each share is set out as follows.

Share	X	Y	Z

Beta	0.50	0.85	1.20

What is the required return for investing in the shares of X, Y and Z?

Question 4.9

The risk free rate has just increased from 5% to 8%.

1. How would this change the intercept point of the capital market line with the efficiency frontier?
2. How would this change the slope of the securities market line?
3. What does this say about the type of securities that would be preferred in times when interest rates are rising?

Question 4.10

Imagine you have R200 000 to invest in a portfolio of two risky securities plus a riskless asset. The following information contains four risky securities that are in different industries:

		Returns					
Outcome	Probability	Share A	Share B	Share C	Share D	Riskless asset	Market return
Depression	20%	-10.0%	10.0%	0.0%	-10.0%	2.0%	-10.0%
Recession	20%	5.0%	10.0%	10.0%	-10.0%	3.5%	-5.0%
Normal	30%	10.0%	5.0%	10.0%	10.0%	6.0%	10.0%
Boom	30%	20.0%	-10.0%	5.0%	25.0%	8.0%	25.0%

Required:

(a) Which two stocks did you choose and why?

(b) Given that your wealth is allocated to these two stocks plus the riskless asset in the proportions of 15%, 45% and 40% respectively, what is your expected rand and percentage return for your portfolio?

(c) Use the Markowitz technique to calculate the variance of this portfolio.

Question 4.11

Mr. Bhamjee places 40% of his funds in Security X which has a return of 12% and 60% of his money in Security Y which has a return of 18%. The standard deviation on Security X is 20% and on Y is 15% respectively.

Required:

(a) Calculate the expected return on the portfolio.

(b) Calculate the standard deviation for the portfolio if the correlation between the stocks is +0.5.

(c) Calculate the standard deviation for the portfolio if the correlation between the stocks is –0.5.

(d) Which of the above correlation factors is the best for the portfolio? Explain why this is so. Page 4-54

Question 4.12

A security analyst provides the following forecasts for the forthcoming year for an investor seeking to invest R100 000:

State of the economy	Probability of state	Return of market portfolio	Return on Security A	Return on government security
Strong	0.2	25%	27%	8%
Moderate	0.5	15%	17%	8%
Weak	0.3	5%	–3%	8%

Required:

(a) Compute the expected return (mean value), standard deviation, and co-efficient of variation for each of the following three portfolios:

(i) The entire amount of R100 000 is invested in the market portfolio.

(ii) An amount of R70 000 is invested in the market portfolio and the remaining R30 000 in a government security.

(iii) An amount of R50 000 is borrowed at the government security rate of 8% and the full R150 000 is invested in the market portfolio.

(b) Plot the above three portfolios on a graph depicting risk ($\boldsymbol{\sigma}_p$) and return (R_p). Please label the relevant aspects clearly.

(c) Critically comment on the investment analysis in part (a) and part (b) above.

(d) Compute the beta for Security A and determine an appropriate return for Security A based on its beta value.

(e) Discuss the significance of the beta value for Security A, computed in part (d) above, in terms of risk and expected return.

Question 4.13

The capital asset pricing model provides a basis for arriving at the expected rate of return on a security or portfolio by reference to the risk-free rate, the expected return on the market, and the beta value of the security or portfolio. The beta value measures only a component of total risk – that which is referred to as systematic risk.

Required:

(a) Using the capital asset pricing model, calculate the expected rate of return on a security for which:

- the risk-free rate is 11%;
- the expected return on the market is 22%; and
- the beta value is 1.2.

(b) Give the reasoning to justify the consideration of only systematic risk in estimating the expected return on a security.

Question 4.14

The following facts pertain to Amtel, a share listed in the industrial sector of the JSE.

Beta co-efficient= 1.6

Expected dividend (D)= R3

Expected growth rate (g) = 5%

In addition it is ascertained that the current risk-free rate is 12% and that the required rate of return on shares with a beta co-efficient of 1 is 17%.

Required:

(a) At what price would you expect Amtel to be quoted today?

(b) Assuming there was an increase in the money supply which reduced the riskless rate to 10%, at what price would you expect Amtel to be quoted?

(c) If, in addition to (b) above, the required rate of return on the market dropped to 15%, at what price would you expect Amtel to be quoted?

(d) In addition to (b) and (c) above, assume new management of Amtel institutes policies that Page 4-55 increase expected growth to 7%, and expected stability in profits causes the beta co-efficient to decline to 1.2. Determine the new equilibrium price.

(e) Discuss two factors that you consider to be significant drawbacks in effectively using the capital asset pricing model for the valuation of shares on the JSE.

Question 4.15

You have recently been appointed as the portfolio manager of a small syndicate holding the following shares:

Share	Investment	Share Beta
P	R800 000	1.2
Q	R400 000	1.8
R	R500 000	2.0
S	R300 000	1.4

It has been established that there is a 60% probability that the market return will be 14%. However, should the recession deepen, the return will drop to 8%.

Required:

(a) Establish the rate of return anticipated from share R if the risk-free rate is 7%.

(b) Calculate the effect on the expected return and risk of the portfolio of the sale of shares in P and reinvestment of the R800 000 in a new share X with a beta of 2.5.

(c) Briefly discuss what action, if any, you would take as the portfolio manager if you anticipate an economic recession in the near future.

Question 4.16

Identify the curves or points referred to below by writing down the relevant letter(s) reflected on the diagram set out on the following page:

(a) the efficient frontier where the investor is able neither to borrow nor to lend;

(b) the efficient frontier where the investor is able to lend at the risk-free rate but is not able to borrow;

(c) the efficient frontier where the investor is able to lend and to borrow at the risk-free rate;

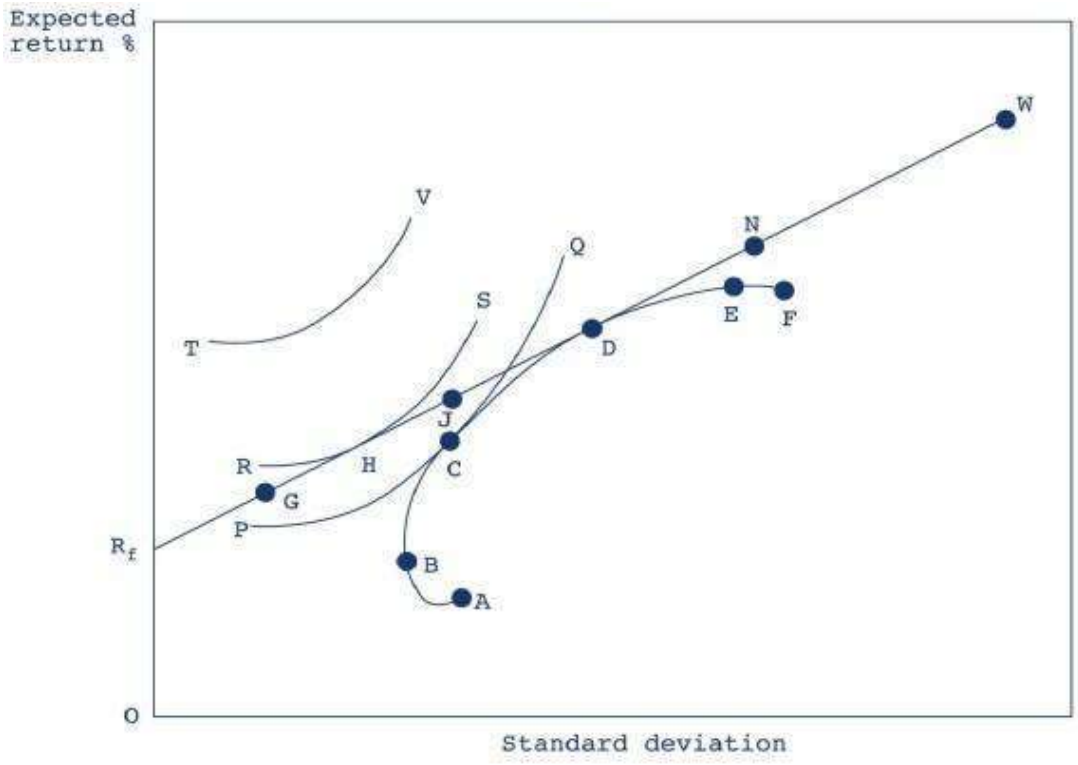
(d) the market portfolio;

(e) a portfolio where 75% of the investor's funds are invested in the market portfolio and 25% in risk-free securities;

(f) an example of a portfolio where the investor has borrowed to invest in the market;

(g) an example of an inefficient portfolio; and

(h) the optimal portfolio for an investor with the set of indifference curves given in the diagram where he is able to invest in risk-free securities.



Page 4-56 **Question 4.17**

The following data are available for four shares listed on the JSE:

	Share	Expected return %	Expected beta
1	Jampak	17.0	1.3
2	Colder	14.5	0.8
3	Moonpak	15.5	1.1
4	Zappi	18.0	1.7

The risk-free rate is 10% and the expected return on a market portfolio is 15%.

Required:

(a) Draw a graph depicting the security-market line and plot each of the four shares.

(b) On the basis of the expectations, identify the shares which are overvalued and those which are undervalued.

(c) If the risk-free rate were to rise to 12% and the expected return on the market portfolio to 16%, which shares, if any, would be undervalued?

(d) Assume you held a portfolio invested equally in each of the four shares under market conditions as per (c) above. You want to invest in a fifth share, so that the portfolio beta is 1.2, and you are equally invested in five shares.

(i) What should be the beta of the fifth share?

(ii) What would be the expected return on the new portfolio?

(iii) Discuss briefly whether it would be appropriate to use beta as a measure of risk for the new portfolio.

Question 4.18

You have been called upon to advise a client with regard to an investment of R100 000 in shares in the industrial sector of the JSE. You have gathered data and assigned probabilities to expected returns under four possible market conditions. The following probabilities have been assigned to two individual shares, and to a unit trust. The unit trust is a widely diversified portfolio with a beta of 1.

Market condition	Probability	% Expected returns		
		Aruntex	Boumet	CD Trust
Poor	0.2	–10	10	–5
Moderate	0.3	5	10	15
Good	0.4	30	15	25
Exceptional	0.1	45	20	35

You may assume that, despite the small number of readings, the distributions all have the characteristics of a normal distribution. In addition, you may assume that borrowing and lending is possible at the risk-free rate of 10%.

Required:

(a) Find the expected return, standard deviation, and co-efficient of variation for each of Aruntex and Boumet.

(b) Find the correlation co-efficient for the returns from Aruntex and Boumet.

(c) Calculate the beta of Boumet.

(d) If your client borrowed R70 000 and invested R170 000 in a certificate of deposit Trust, calculate the expected return and total risk of the investment. Page 4-57

(e) If your client invested R20 000 in Aruntex and R80 000 in Boumet, calculate the expected return and total risk of the portfolio.

(f) Identify three simplifying assumptions used in developing the capital asset pricing model.

Question 4.19

The following information gives estimated data concerning share returns:

State of the economy	Probability of each state occurring	Rates of return if state occurs	
		Market	Share A
Deep recession	0.05	–20%	–30%
Mild recession	0.25	10	5
Average	0.35	15	20

Mild boom	0.20	20	25
Strong boom	0.15	25	30

In addition, it is estimated that the risk-free rate will be constant at 8%.

Required:

(a) Calculate the expected rates of return on the market and share A.

(b) Calculate the estimated beta for share A. What is the market's estimated beta?

(c) Based on the estimated (*ex ante*) beta, calculate the required rate of return for share A. Would it be more meaningful to use an *ex post* beta to calculate the required rate of return? Explain your answer.

Question 4.20

Helga Mnguni has recently graduated. During her three years of study she has accumulated savings of R10 000 and a good grasp of the theories underlying risk and return as they apply to the JSE. She has followed three particular companies closely and has used statistical data relating to these shares in a project. She now wishes to invest her funds into some combination of these shares applying the principles of portfolio theory. Her data for the three companies are as follows:

	Barloworld	**Delta**	**Wooltru**
Expected return	15%	25%	20%
Variance of return	64%	225%	81%
Beta	1.0	1.5	1.2

	Correlation co-efficient
Barloworld/Delta	–0.3
Barloworld/Wooltru	+0.6
Delta/Wooltru	+0.2

Helga wishes to invest in a portfolio which comprises two of the three shares, applying R5 000 to each of the two selected.

Required:

(a) Determine the portfolio of two shares which would be most attractive and comment on your recommendation.

(b) Calculate the covariance of returns on the market with returns of Delta.

(c) Identify the fundamental principle of portfolio theory and the most significant assumptions on which portfolio theory is based.

Question 4.21

Two shares record the following percentage returns based on five readings:

	1	2	3	4	5
Share A	16%	10%	4%	18%	12%
Share B	22%	40%	20%	-10%	28%
Market	20%	32%	14%	-6%	20%

Required:

(a) Calculate the correlation co-efficient for shares A and B.

(b) Calculate the beta of B.

(c) Sketch the possible combinations of A and B which could be held by an investor in a risk/return space. Label both axes and Points A and B. Indicate the following on the graph with regard to the portfolio:

(i) Point P, where an investor who is highly risk-averse may invest.

(ii) Point Q, where an investor who is not highly risk-averse may invest.

(iii) Point R, where an investor could, but would not, invest.

(d) If the risk-free rate is 9%, and the return on the market is 11.5% sketch the securities market line. Label both axes clearly. Indicate the following on the graph:

(i) Point X, a share with the same systematic risk as the market.

(ii) Point Y, an overpriced share.

(iii) Point Z, underpriced share.

Question 4.22

Using the letters of the alphabet given below, label the following clearly on a diagram of the security market line where the risk-free rate is 15%:

(a) the systematic risk and the expected rate of return on the market portfolio;

(b) an example of an underpriced security;

(c) an example of an overpriced security;

(d) an example of a correctly priced security;

(e) the systematic risk and the expected rate of return for a portfolio where 25% of the total resources are invested in risk-free securities and 75% in the market portfolio;

(f) the expected rate of return and systematic risk for an example of a security constituting a defensive investment; and

(g) the expected rate of return and systematic risk for an example of a security constituting an aggressive investment.

Question 4.23

Describe the three concepts of the 'efficient market hypothesis' (EMH) and discuss the possible implications of these concepts with respect to technical analysis and fundamental analysis.

Question 4.24

You are required to discuss why investors' utility curves are important in modern portfolio theory (MPT). Use a diagram (graph) to demonstrate your answer.

Question 4.25

Studies have indicated that a firm's annual financial statements are an important source of information for making equity investment decisions. Yet, other studies indicate that share prices do not react significantly to the publication of the annual financial statements. How would you reconcile these results with each other and with the Efficient Market Hypothesis?

---

[1] In order to be accurate, we should really divide by $(n - 1)$ rather than n in our examples. See for example, Table 4.4 and the Self-study example at the end of the chapter in which we should divide the squared deviations by $(5 - 1)$ rather than by 5. In practice, we will be working with very large samples and so there will be very little difference between the use of $(n - 1)$ and n. So we have retained n to reflect the reality that the results will be very close in practice whilst in our examples which have very limited observations, the effect will be significant. In Excel, use the =STDEV function to reflect the standard deviation of a sample.

[2] Van Rensburg, P. & Robertson, M. (2003) Size, price-to-earnings and beta on the JSE, *Investment Analyst Journal*, 58, 7-16.

[3] McKane, G. & Britten, J. (2018) Liquidity and size effects on the JSE, *Investment Analysts Journal*, 47 (3) 229-242.

[4] Wessels, DR and Krige, JD (2005). The persistence of active fund management performance, *SA Journal of Business Management*, 36(2).

[5] Bradfield, D and Swartz, J (2001). Recent evidence on the persistence of fund performance – a note. *SA Journal of Accounting Research*, 15(2).

[6] In recent years, Allan Gray has found it more difficult to outperform the market benchmark for some of its funds. For example, the performance of the Allan Gray SA Equity Fund was 6.7% per year since inception in 2015 as compared to the market benchmark of 8.6%. This may reflect Allan Gray's focus on value and contrarian investing as well as the state of the market over the last decade.

[7] You can buy gold coins such as Krugerrands but trading costs are high and there are insurance costs and storage costs, although it may be reassuring to touch the gold you own.

[8] The standard deviation of monthly returns from 2008 to 2023 for the NewGold ETF was 5.6% whilst for Gold Fields it was 14.1%, for AngloGold it was 14.5% and for Harmony it was 18.9%. The compound annualised growth rates were as follows: NewGold ETF 10.7%, Gold Fields 7.4%, AngloGold 2.6%, Harmony -0.9%, Portfolio of the 3 mining companies, 3.9%.The NewGold ETF offered a higher return at a lower risk over this period. The compound annualised growth rates until 2018 were as follows: NewGold 8.4%, Gold Fields -8.9%, AngloGold -7.6% and Harmony -13%. When we use arithmetic means until 2018, we find that Harmony had a positive average return as well as AngloGold, which indicates the effect that volatility can have on arithmetic means. Rather focus on the compound annual returns. (See Chapter 3 for more on arithmetic and geometric (compound) returns).

[9] Exchange Traded Notes (ETNs) are debentures issued by a bank in which the return is directly linked to an index or commodity. This implies greater risk as it is based on a promise by the bank. What if the bank fails? Although NewGold has been structured in the form of debentures, there is a separate special purpose company which issues the debentures and which is independent of ABSA and therefore will not be affected if ABSA fails. In substance, NewGold is an ETF.

[10] Introduced through the following articles: (i) Sharpe, W., "Capital Asset Prices: A Theory of Market Equilibrium", *Journal of Finance*, 1964; (ii) Lintner, J., "The Valuation of Risk Assets and the Selection of Risky Investments in Stock Portfolios and Capital Budgets", *Review of Economics and Statistics*, 1965; and (iii) Mossin, J., "Equilibrium in a Capital Asset Market", *Econometrica*, 1966.

[11] This is based on the article, **On estimating the beta coefficient**, by D. Bradfield in the *Investment Analyst Journal*, No. 5, 2003, pp.47-52.

[12] Size of final retirement benefit versus the required level of income (from that benefit).

[13] Co-founder of Oaktree Capital Management.

[14] Portfolio managers will often point to *capital preservation* as a key part of their investing philosophy because they understand how more important protecting capital and avoiding losses is for investors. Portfolio allocations will include insurance type investments to avoid losses such as cash, money market investments and bonds. Equity investments can be in such defensive sectors such as property and healthcare and high-risk equity investments will include venture capital, biotech and mining exploration companies. Most investors will only allocate a very small proportion of their funds to these high-risk equity investments.